

3

MONTE CARLO INTEGRATION

In this section, we introduce Monte Carlo integration and importance sampling in detail. We will show how these ideas can be applied to a variety of problems such as computing integrals, computing expectations, sampling from complex distributions, and computing marginal likelihoods.



3.1 INTRODUCTION TO MONTE CARLO INTEGRATION

We have repeatedly highlighted that we are interested in sampling from a probability measure p_* . One reason we are interested in this is to *estimate expectations* of certain measures, i.e., we can estimate moments of distributions. Of course, so far, we have been considering drawing samples from known distributions (for which moments might be readily available). However, it is often the case in sampling applications that, in most cases, the primary goal *is* to compute expectations for distributions which are not available to us in closed form.

We will call this task as *Monte Carlo integration*. Briefly, given a probability distribution p_* , we are interested in computing expectations of the form

$$\bar{\varphi} = \mathbb{E}_{p_*}[\varphi(X)] = \int \varphi(x)p_*(x)dx = (\varphi, p_*),$$

where $\varphi(x)$ is called a *test function*. For example, $\varphi(x) = x$ would give us the mean, $\varphi(x) = x^2$ the second moment, or $\varphi(x) = \log(x)$ would give us the entropy. For example, given $X^{(1)}, \dots, X^{(N)} \sim p_*$ i.i.d, we know that (intuitively, at this point) the mean estimator is given by

$$\mathbb{E}_{p_*}[X] = \int xp_*(x)dx \approx \frac{1}{N} \sum_{i=1}^N X^{(i)},$$

which is simply the empirical average of the samples. While this can be intuitive, it underlies a certain choice about the approximation of the probability measure $p_*(dx) := p_*(x)dx$ using its

samples. In order to do this, we build an *empirical distribution* of the samples, using

$$p_{\star}^N(dx) = \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}}(dx). \quad (3.1)$$

In order to understand how this works, we first need to understand the Dirac delta measure δ_x . The Dirac delta measure is defined as

$$f(y) = \int f(x) \delta_y(dx) \quad (3.2)$$

Here, the Dirac can be thought as a *point mass* at y . In other words, the Dirac delta measure is a measure which is concentrated at a single point. To understand it intuitively, the object $\delta_y(x)$ can be informally thought as a function centered at y (and only takes value 1 at y)¹

$$\delta_y(x) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise.} \end{cases}$$

One can see that then $p_{\star}^N$ is a *sample based approximation* of $p_{\star}$, where the samples are equally weighted. While we never may use this particular approximation of as a density approximation, it is useful to build estimates of expectations. Generalising the above scenario, let us consider the estimation of the general expectation

$$\bar{\varphi} = (\varphi, p_{\star}) = \mathbb{E}_{p_{\star}}[\varphi(X)] = \int \varphi(x) p_{\star}(x) dx.$$

Given samples $X^{(1)}, \dots, X^{(N)}$, we can build $p_{\star}^N$ as in (3.1) and approximate this expectation as

$$\begin{aligned} \bar{\varphi} &= \mathbb{E}_{p_{\star}}[\varphi(x)] = (\varphi, p_{\star}) \\ &= \int \varphi(x) p_{\star}(x) dx \\ &\approx \int \varphi(x) p_{\star}^N(dx) = (\varphi, p_{\star}^N) \\ &= \int \varphi(x) \frac{1}{N} \sum_{i=1}^N \delta_{X_i}(dx) \\ &= \frac{1}{N} \sum_{i=1}^N \int \varphi(x) \delta_{X_i}(dx) \\ &= \frac{1}{N} \sum_{i=1}^N \varphi(X_i) = \hat{\varphi}_{\text{MC}}^N, \end{aligned} \quad (3.3)$$

where we have used (3.2) in the approximate integral to arrive at the final expression. Note that this generalises the example above about the mean (which was $\varphi(x) = x$ case). In this course, we will also be interested in the properties of these estimators.

¹This is not correct rigorously – just for intuition! Note that the Diracs always make sense with an integral attached to them.

Remark 3.1. As we can see that, the Monte Carlo estimator can be used to estimate expectations. We can also use this idea to estimate integrals. Consider a standard integration problem

$$I = \int f(x)dx,$$

where $f(x)$ is a function. We can use the Monte Carlo (MC) estimator to estimate this integral as

$$\begin{aligned} I &= \int \frac{f(x)}{p(x)} p(x) dx \\ &\approx \int \frac{f(x)}{p(x)} p^N(dx) \quad \text{where } p^N(dx) = \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}}(dx) \\ &= \frac{1}{N} \sum_{i=1}^N \frac{f(X_i)}{p(X_i)}. \quad \text{using (3.1)} \end{aligned}$$

In this case, we have $\varphi(x) = \frac{f(x)}{p(x)}$. This is particularly easy for the integrals of type

$$I = \int_0^1 f(x)dx,$$

where $f(x)$ is a function. In this case, we can use the uniform distribution as the base distribution p and use the Monte Carlo estimator to estimate the integral without needing to compute any ratios.

In the following, we prove some results about the properties of the Monte Carlo estimator (3.3) when samples are i.i.d from p_* .

Proposition 3.1. *Let $X_1, \dots, X_N$ be i.i.d samples from p_* . Then, the Monte Carlo estimator*

$$\hat{\varphi}_{\text{MC}}^N = \frac{1}{N} \sum_{i=1}^N \varphi(X_i)$$

is unbiased, i.e.,

$$\mathbb{E}_{p_*}[\hat{\varphi}_{\text{MC}}^N] = \bar{\varphi}.$$

Proof. We have

$$\begin{aligned}
\mathbb{E}_{p_\star}[\hat{\varphi}_{\text{MC}}^N] &= \mathbb{E}_{p_\star} \left[\frac{1}{N} \sum_{i=1}^N \varphi(X_i) \right] \\
&= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{p_\star}[\varphi(X_i)] \\
&= \frac{1}{N} \sum_{i=1}^N \int \varphi(x) p_\star(x) dx \\
&= \int \varphi(x) p_\star(x) dx \\
&= \bar{\varphi},
\end{aligned}$$

which proves the result. $\square$

Next, we can also compute the variance of the Monte Carlo estimator.

Proposition 3.2. *Let $X_1, \dots, X_N$ be i.i.d samples from $p_\star$. Then, the Monte Carlo estimator*

$$\hat{\varphi}_{\text{MC}}^N = \frac{1}{N} \sum_{i=1}^N \varphi(X_i)$$

has variance

$$\text{var}_{p_\star}[\hat{\varphi}_{\text{MC}}^N] = \frac{1}{N} \text{var}_{p_\star}[\varphi(X)],$$

where

$$\text{var}_{p_\star}[\varphi(X)] = \int (\varphi(x) - \bar{\varphi})^2 p_\star(x) dx.$$

Proof. We have

$$\begin{aligned}
\text{var}_{p_\star}[\hat{\varphi}_{\text{MC}}^N] &= \text{var}_{p_\star} \left[\frac{1}{N} \sum_{i=1}^N \varphi(X_i) \right] \\
&= \frac{1}{N^2} \sum_{i=1}^N \text{var}_{p_\star}[\varphi(X_i)] \\
&= \frac{1}{N^2} \sum_{i=1}^N \int (\varphi(x) - \bar{\varphi})^2 p_\star(x) dx \\
&= \frac{1}{N} \text{var}_{p_\star}[\varphi(X)] = \frac{\sigma_\varphi^2}{N}
\end{aligned}$$

Provided that $\text{var}_{p_\star}[\varphi(X)] < \infty$ and the estimator is consistent as $N \rightarrow \infty$. This proves the result. $\square$

Remark 3.2. The expression $\text{var}_{p_*}[\hat{\varphi}_{\text{MC}}^N]$ is the variance of the MC estimator but this expression requires the true mean $\bar{\varphi}$ to be known. In practice, we do not know the true mean but also have an MC estimator for it. We can plug this estimator into the variance in order to obtain an empirical variance estimator. Note that

$$\begin{aligned}\text{var}_{p_*}[\hat{\varphi}_{\text{MC}}^N] &= \frac{1}{N} \text{var}_{p_*}[\varphi(X)] \\ &= \frac{1}{N} \int (\varphi(x) - \bar{\varphi})^2 p_*(x) dx \\ &\approx \frac{1}{N^2} \sum_{i=1}^N (\varphi(X_i) - \hat{\varphi}_{\text{MC}}^N)^2 \\ &= \sigma_{\varphi, N}^2.\end{aligned}$$

This estimator then can be used to estimate the variance of the MC estimator.

We can therefore obtain a central limit theorem for our estimator, i.e.,

$$\frac{(\hat{\varphi}_{\text{MC}}^N - \bar{\varphi})}{\sigma_{\varphi, N}} \rightarrow \mathcal{N}(0, 1) \quad \text{as} \quad N \rightarrow \infty.$$

This can be used to build empirical confidence intervals for the estimators. However, this is not a principled estimate and may not be valid in many scenarios. We can also see that we have a standard deviation estimate (which follows from the variance estimate) given by

$$\text{std}_{p_*}[\hat{\varphi}_{\text{MC}}^N] = \sqrt{\text{var}_{p_*}[\hat{\varphi}_{\text{MC}}^N]} = \frac{\sigma_{\varphi}}{\sqrt{N}}.$$

This is a typical display of a convergence rate $\mathcal{O}(1/\sqrt{N})$.

Remark 3.3. One of the most common application of sampling is to estimate probabilities. We have seen that different choices of φ can lead to estimating different quantities such as the mean and n th moments. However, the MC estimators can also be used to estimate probabilities. In order to see this, assume that we would like to estimate $\mathbb{P}(X \in A)$ where $X \sim p_*$. We know that this is given as

$$\mathbb{P}(X \in A) = \int_A p_*(x) dx,$$

see, e.g., Definition ???. For example, A can simply be an interval. Given the definition above, we can write

$$\begin{aligned}\mathbb{P}(X \in A) &= \int_A p_*(x) dx \\ &= \int \mathbf{1}_A(x) p_*(x) dx,\end{aligned}$$

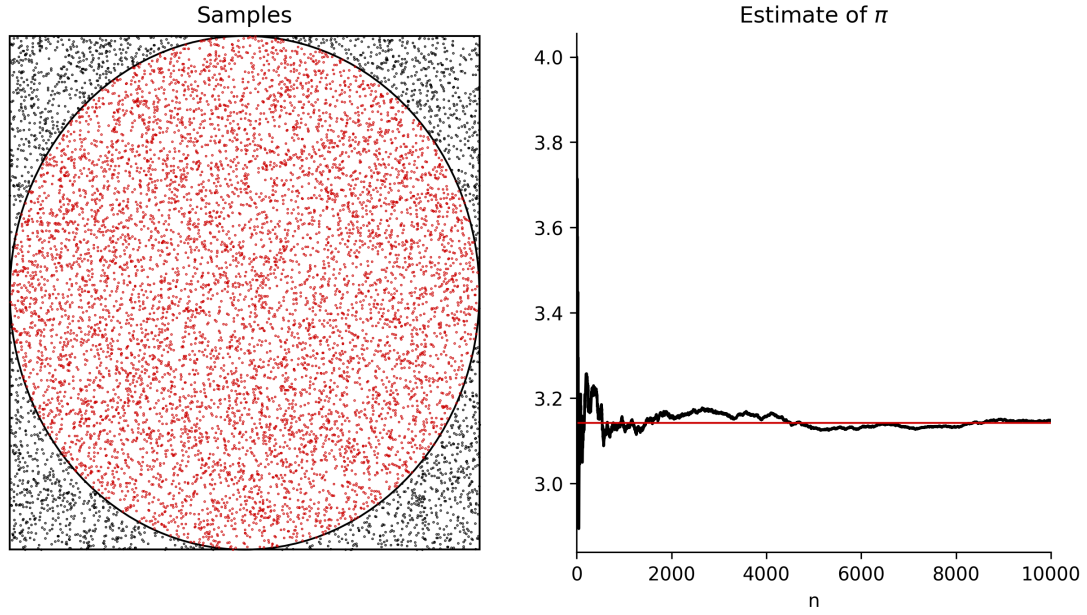


Figure 3.1: Estimating π using the Monte Carlo method.

where $\mathbf{1}_A(x)$ is the indicator function of A . We can therefore set $\varphi(x) = \mathbf{1}_A(x)$ and given the samples from p_* , we can build an estimator

$$\begin{aligned}
 \mathbb{P}(X \in A) &= \int \mathbf{1}_A(x) p_*(x) dx, \\
 &\approx \int \mathbf{1}_A(x) p_*^N(dx), \\
 &= \int \mathbf{1}_A(x) \frac{1}{N} \sum_{i=1}^N \delta_{X_i}(dx), \\
 &= \frac{1}{N} \sum_{i=1}^N \int \mathbf{1}_A(x) \delta_{X_i}(dx), \\
 &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}_A(X_i).
 \end{aligned}$$

This estimator also leads to an intuitive procedure: We sample $X_1, \dots, X_N$ from p_* and we effectively just count the samples in A and divide it by N .

We can now return to the example of estimating π using the Monte Carlo method.

Example 3.1. Recall the problem of estimating π using the Monte Carlo method. Frame it as a Monte Carlo integration problem and derive the algorithm rigorously.

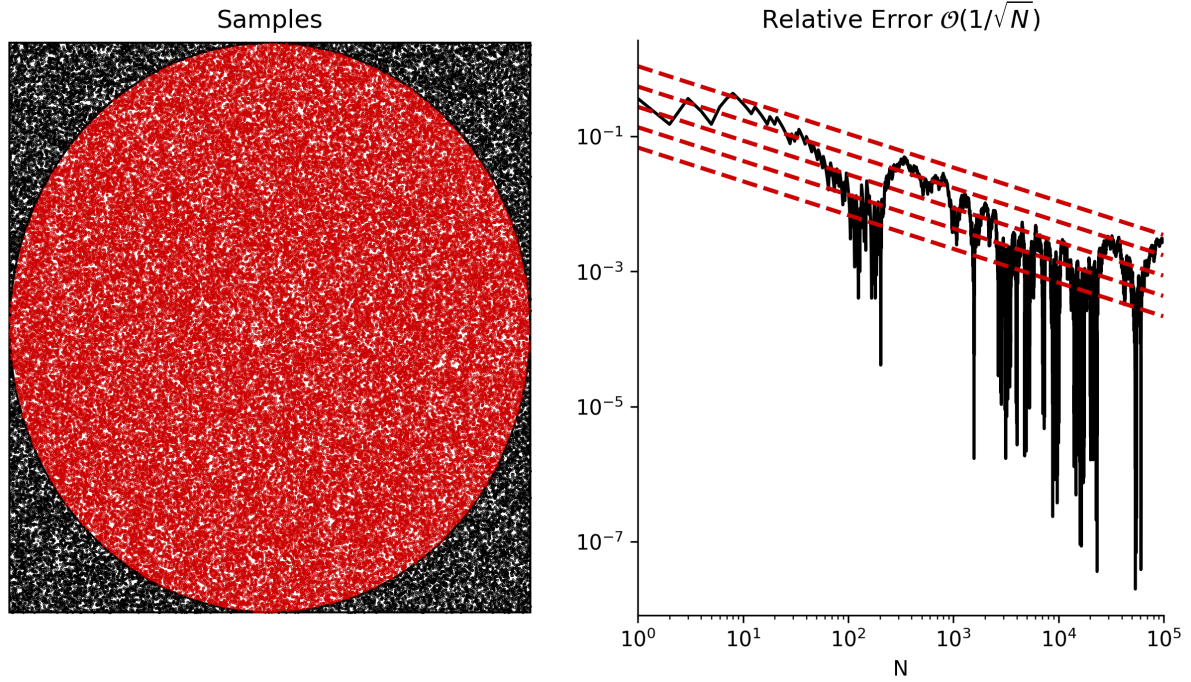


Figure 3.2: Relative error (see next section) of the Monte Carlo estimate provided by sampling within the disk.

Solution. The logic that was used in this example was to estimate the area of a disk that lies within a square. To be precise, consider the square $[-1, 1] \times [-1, 1]$ and define the uniform distribution on this square as $p(x, y) = \text{Unif}([-1, 1] \times [-1, 1])$. We can simply phrase the problem as estimating the area of the disk which we define as $A \subset [-1, 1] \times [-1, 1]$. The set A is given as

$$A = \{(x, y) \in [-1, 1] \times [-1, 1] \quad \text{s.t.} \quad x^2 + y^2 \leq 1\}.$$

We can then formalise this problem as estimating the probability that a point lies within the disk. This is given as

$$\begin{aligned} \mathbb{P}(X \in A) &= \int_A p_*(x, y) dx dy, \\ &= \int \mathbf{1}_A(x, y) p_*(x, y) dx dy. \end{aligned}$$

Sampling $(X_i, Y_i) \sim p_*(x, y)$ (a uniform sample within a square), we can estimate this integral using the standard MC method. More formally, we can write

$$\begin{aligned} \mathbb{P}(A) &= \int_A p_*(x, y) dx dy \\ &= \int \mathbf{1}_A(x, y) p_*(x, y) dx dy, \\ &\approx \frac{1}{N} \sum_{i=1}^N \mathbf{1}_A(X_i, Y_i) \rightarrow \frac{\pi}{4} \quad \text{as } N \rightarrow \infty. \end{aligned}$$

A trajectory of the estimation procedure π can be seen from Fig. 3.1 w.r.t. varying sample size.

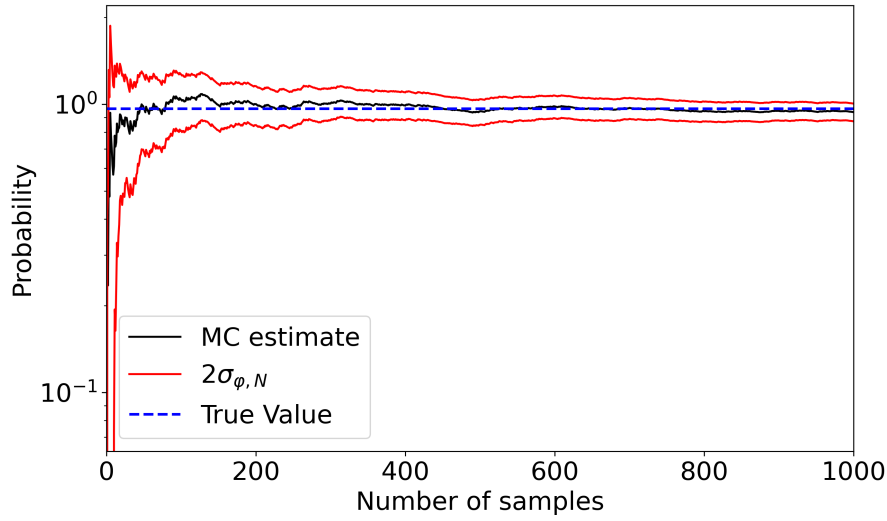


Figure 3.3: Monte Carlo integration of $h(x) = [\cos(50x) + \sin(20x)]^2$

Nonasymptotic results showing the convergence rate of $\mathcal{O}(1/\sqrt{N})$ are also available (see, e.g., [Akyildiz \(2019, Corollary 2.1\)](#)) – see Fig. 3.2 for a demonstration.

Example 3.2 (Example 3.4 from [Robert and Casella \(2004\)](#)). Let us consider an example of estimating an integral:

$$I = \int_0^1 h(x)dx = \int_0^1 [\cos(50x) + \sin(20x)]^2 dx.$$

The exact value of this integral is 0.965. Describe a Monte Carlo method to estimate this integral.

Solution. We can just choose $p(x) = \text{Unif}(0, 1)$ and set $\varphi(x) = h(x)$. We can then write

$$\begin{aligned} I &= \int_0^1 h(x)dx, \\ &= \int_0^1 \varphi(x)p(x)dx, \end{aligned}$$

and apply the standard MC estimator. The results (together with the empirical variance estimate) can be seen from Fig. 3.3.

Finally, we provide an example of estimating the probability of a random variable.

Example 3.3. Consider $X \sim \mathcal{N}(0, 1)$ and we would like to estimate the probability that $X > 2$. Describe the MC method.

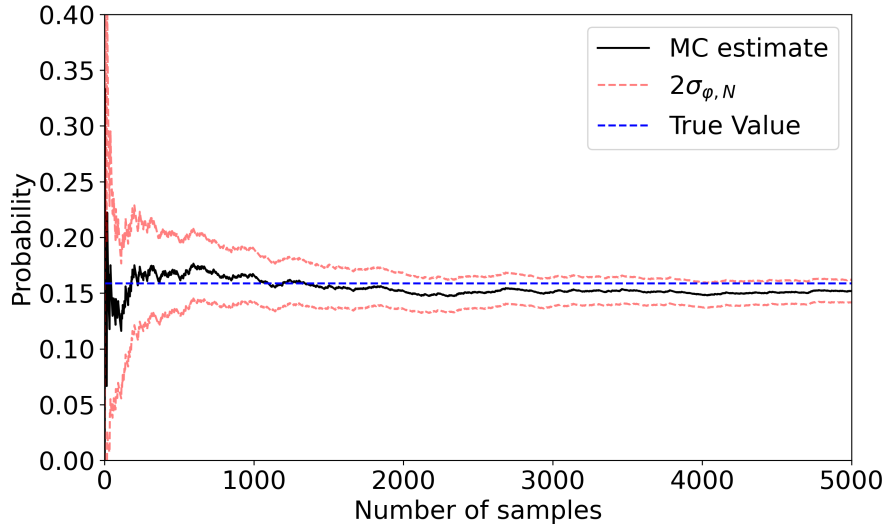


Figure 3.4: Monte Carlo estimation of the tail probability $X > 2$. The “true value” is computed via numerical integration.

Solution. The way to do this is to choose

$$p(x) = \mathcal{N}(0, 1), \quad \varphi(x) = \mathbf{1}_{\{x > 2\}}(x).$$

We can then write that

$$\begin{aligned} \mathbb{P}(X > 2) &= \int_{-\infty}^{\infty} \mathbf{1}_{\{x > 2\}}(x) \mathcal{N}(0, 1) dx, \\ &\approx \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{X_i > 2\}}(X_i). \end{aligned}$$

where $X_1, \dots, X_N \sim \mathcal{N}(0, 1)$. The results can be seen from Fig. 3.4.

Exercise 3.1. Suppose given a distribution $p_X(x)$, we would like to estimate its cumulative distribution function (CDF) $F_X(x)$. Recall that pointwise, the CDF is defined as

$$F_X(x) = \int_{-\infty}^x p_X(s) ds.$$

Convert this problem to a Monte Carlo integration problem by suitably identifying the test function. Provide the Monte Carlo estimator by using N samples $X_1, \dots, X_N$ from p_X (called empirical distribution function) for fixed x . Compute its variance in terms of $F_X(x)$.

Exercise 3.2. (Kernel density estimation) Consider a set of samples $x_i \sim p(x)$, $i = 1, \dots, N$. We would like to estimate the density $p(x)$ using kernel density estimation. The kernel density estimate is in general given by

$$\hat{p}(x) = \frac{1}{N} \sum_{i=1}^N K_h(x - x_i) = \frac{1}{Nh} \sum_{i=1}^N K\left(\frac{x - x_i}{h}\right),$$

for some kernel K and bandwidth h . Show that, for fixed x , the KDE $\hat{p}(x)$ is a Monte Carlo estimator. What is the test function?

Exercise 3.3. Consider the following integral:

$$I = \int_0^1 (1 - x^2)^{1/2} dx.$$

Take the sampling distribution as uniform on $[0, 1]$. Build the Monte Carlo estimate for varying N and compute the mean and the variance, i.e., for $\varphi(x) = (1 - x^2)^{1/2}$ and given $X_1, \dots, X_N$ from $\text{Unif}(0, 1)$, compute

$$\hat{\varphi}^N = \frac{1}{N} \sum_{i=1}^N \varphi(X_i) \tag{3.4}$$

and the empirical variance

$$\widehat{\text{var}}[\hat{\varphi}^N] = \frac{1}{N^2} \sum_{i=1}^N (\varphi(X_i) - \hat{\varphi}^N)^2. \tag{3.5}$$

Discuss the difference between this empirical estimator vs. the correct one for the variance (using the true value). Plot your mean estimates (3.4), standard deviation estimates (by taking the square root of (3.5)), and the true value $I = \pi/4$. Can you always trust the variance estimates?

Exercise 3.4. It is often stated in textbooks that the variance (hence the MSE) of the MC estimator derived in Proposition 3.2 does not depend on the dimension of the state-space. We will examine this claim below (also see Branchini (2024) for a fun blogpost covering one of the examples below as well as commentary about the central texts where this claim appears).

Consider the multivariate Gaussian target distribution

$$p(x) = \mathcal{N}(x; 0, I_d),$$

where I_d is the $d \times d$ identity matrix. Consider the following test functions:

(i)

$$\varphi(x) = e^{-\|x\|^2/4}.$$

(ii)

$$\varphi(x) = \|x\|^2/2$$

(iii)

$$\varphi(x) = e^{\|x\|^2/8}.$$

For each example, compute the variance of the MC estimators as given in Proposition 3.2 as a function of the dimension d .

3.2 ERROR METRICS

In order to quantify convergence, we will have a number of error metrics in the following section. Let us generically denote $\hat{\varphi}^N$ as an estimator of $\bar{\varphi}$ within this section – which can be standard Monte Carlo estimator or any other estimator we will introduce later.

We start with defining the bias as

$$\text{bias}(\hat{\varphi}^N) = \mathbb{E}[\hat{\varphi}^N] - \bar{\varphi}, \quad (3.6)$$

where $\bar{\varphi}$ is the *true value*. We call an estimator *unbiased* if the bias is zero. In the case where we sample i.i.d from $p(x)$, we can build unbiased estimators of expectations and integrals. We recall the variance

$$\text{var}(\hat{\varphi}^N) = \mathbb{E}[(\hat{\varphi}^N - \mathbb{E}[\hat{\varphi}^N])^2]. \quad (3.7)$$

If the estimator is unbiased, we can then replace $\mathbb{E}[\hat{\varphi}^N]$ with $\bar{\varphi}$. Next, we define the mean squared error (MSE)

$$\text{MSE}(\hat{\varphi}^N) = \mathbb{E}[(\hat{\varphi}^N - \bar{\varphi})^2]. \quad (3.8)$$

One can see that the MSE and the variance coincides if the estimator is unbiased. We have also the following decomposition of the MSE

$$\text{MSE}(\hat{\varphi}^N) = \text{bias}(\hat{\varphi}^N)^2 + \text{var}(\hat{\varphi}^N). \quad (3.9)$$

We define the root mean square error (RMSE) as

$$\text{RMSE}(\hat{\varphi}^N) = \sqrt{\text{MSE}(\hat{\varphi}^N)}. \quad (3.10)$$

Finally, we define the relative absolute error (RAE) as

$$\text{RAE}(\hat{\varphi}^N) = \frac{|\hat{\varphi}^N - \bar{\varphi}|}{|\bar{\varphi}|}. \quad (3.11)$$

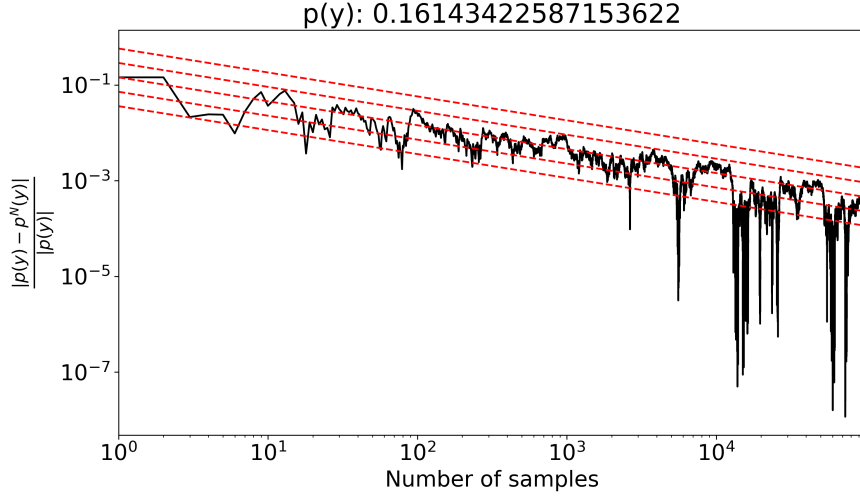


Figure 3.5: Estimating the marginal likelihood $p(y)$ for $y = 1$. One can clearly see the displayed error rate that is $\mathcal{O}(1/\sqrt{N})$.

We usually plot the absolute error of the estimator, as we only run the experiment once in general². We note that this absolute error $|\hat{\varphi}^N - \bar{\varphi}|$ is a random variable (since no expectations are taken). However, this quantity provably converges with a rate of $\mathcal{O}(1/\sqrt{N})$ (see, e.g., [Akyildiz \(2019, Corollary 2.1\)](#)) for the standard Monte Carlo estimator. More precisely, we can write

$$|\hat{\varphi}_{\text{MC}}^N - \bar{\varphi}| \leq \frac{V}{\sqrt{N}}, \quad (3.12)$$

where V is an almost surely finite random variable. This error rate will be displayed empirically in the following sections (see also Fig . 3.2).

Example 3.4 (Marginal likelihood estimation). Recall that, given a prior $p_X(x)$ and a likelihood $p_{Y|X}(y|x)$, we can compute the marginal likelihood $p_Y(y)$ as

$$p_Y(y) = \int p_{Y|X}(y|x)p_X(x)dx.$$

Describe the MC method to estimate $p_Y(y)$.

Solution. This defines a nice integration problem that we can solve using MC. Assume that we are given the following model

$$\begin{aligned} p_X(x) &= \mathcal{N}(x; \mu_0, \sigma_0^2), \\ p_{Y|X}(y|x) &= \mathcal{N}(y; x, \sigma^2). \end{aligned}$$

Assume that $\mu_0 = 0$, $\sigma_0 = 1$, $\sigma = 2$, and $y = 1$. For example, for fixed $y = 1$, we would want to estimate $p_Y(y)$. This integral becomes

$$p_Y(y) = \int p_{Y|X}(y|x)p_X(x)dx,$$

²However, if you were to do a proper experimentation, then you would have to run the same experiment M times (Monte Carlo runs) and average the error to estimate the RMSE.

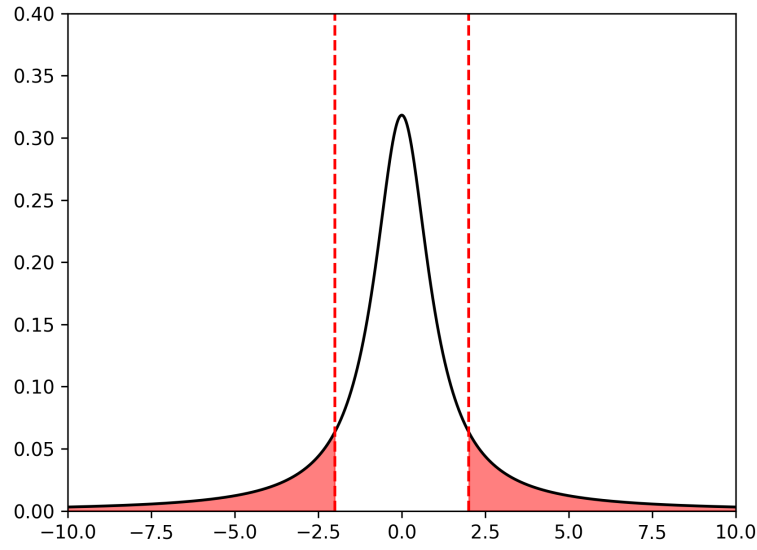


Figure 3.6: Cauchy density of Example 3.5.

where we can set $\varphi(x) = p_{Y|X}(y|x)$. We can then compute the integral using MC estimation procedure as

$$p_Y^N(y) = \frac{1}{N} \sum_{i=1}^N p_{Y|X}(y|X_i),$$

where $X_1, \dots, X_N \sim p_X(x)$. The results can be seen from Fig. 3.5.

This naive approach does not work well in high dimensions as the variance of the estimator can be very large. We will see later how to improve this estimator using importance sampling.

We will next consider another example of estimating a probability where we show how to quantify the variance using the true value.

Example 3.5. Consider the following density

$$p(x) = \frac{1}{\pi(1+x^2)}.$$

We would like to compute the probability of $X \sim p(x)$ being larger than 2, i.e., $\mathbb{P}(X > 2)$. Describe the method and suggest an improvement.

Solution. We can compute this probability using MC estimation as we can set

$$\varphi(x) = \mathbf{1}_{\{x>2\}}(x).$$

We can compute

$$\begin{aligned}\mathbb{P}(X > 2) &= \int_2^\infty p(x)dx \\ &= \int \mathbf{1}_{\{x>2\}}(x)p(x)dx.\end{aligned}$$

We can also compute the real value of this integral as (see Example 2.3 for the CDF of this density)

$$I = \bar{\varphi} = \int_2^\infty p(x)dx = F_X(\infty) - F_X(2) = \frac{1}{2} - \frac{1}{\pi} \tan^{-1}(2) = 0.1476.$$

Let us compute the variance of the Monte Carlo estimator for $N = 10$ samples:

$$\text{var}(\hat{\varphi}_{\text{MC}}^N) = \frac{\text{var}_p(\varphi)}{N}$$

So we need to compute:

$$\begin{aligned}\text{var}_p(\varphi) &= \int \varphi(x)^2 p(x)dx - \left(\int \varphi(x)p(x)dx \right)^2 \\ &= \int \mathbf{1}_{\{x>2\}}(x)^2 p(x)dx - \left(\int \mathbf{1}_{\{x>2\}}(x)p(x)dx \right)^2 \\ &= \int \mathbf{1}_{\{x>2\}}(x)p(x)dx - \left(\int \mathbf{1}_{\{x>2\}}(x)p(x)dx \right)^2 \\ &= 0.1476 - 0.1476^2 = 0.125.\end{aligned}$$

The variance of the estimator then

$$\text{var}(\hat{\varphi}_{\text{MC}}^N) = \frac{0.125}{10} = 0.0125.$$

Could we do better? An idea is to use the fact that the density is symmetric around zero: This means $P(X > 2) = P(X < -2)$ (see Fig. 3.6). So we could compute:

$$\mathbb{P}(|X| > 2) = \mathbb{P}(X > 2) + \mathbb{P}(X < -2) = 2I.$$

Therefore, our new problem is $I = \frac{1}{2}\mathbb{P}(|X| > 2)$. Let us write it as

$$\begin{aligned}I &= \frac{1}{2} \int_{|x|>2} p(x)dx, \\ &= \int \frac{1}{2} \mathbf{1}_{\{|x|>2\}}(x)p(x)dx,\end{aligned}$$

Now define the test function

$$\varphi(x) = \frac{1}{2} \mathbf{1}_{\{|x|>2\}}(x).$$

As before, we need to compute $\text{var}_p(\varphi)$:

$$\begin{aligned}\text{var}_p(\varphi) &= \int \varphi(x)^2 p(x) dx - \left(\int \varphi(x) p(x) dx \right)^2 \\ &= \int \frac{1}{4} \mathbf{1}_{\{|x|>2\}}^2 p(x) dx - \left(\int \frac{1}{2} \mathbf{1}_{\{|x|>2\}} p(x) dx \right)^2 \\ &= \int \frac{1}{4} \mathbf{1}_{\{|x|>2\}} p(x) dx - \left(\int \frac{1}{2} \mathbf{1}_{\{|x|>2\}} p(x) dx \right)^2 \\ &= \frac{1}{4} \times 2 \times 0.1476 - \frac{1}{4} \times (2 \times 0.1476)^2, \\ &= 0.052.\end{aligned}$$

Therefore, the variance of the estimator for $N = 10$ samples is

$$\text{var}(\hat{\varphi}_{\text{MC}}^N) = \frac{0.052}{10} = 0.0052.$$

Improvement over the previous estimator! This kind of variance improvements are crucial in safety critical applications.

Exercise 3.5. We have seen two ways to compute $\mathbb{P}(X > 2)$ for a Cauchy r.v. X with density

$$p(x) = \frac{1}{\pi(1+x^2)}.$$

Now we will explore a third way.

- Express the integral we want to compute, that is $\mathbb{P}(X > 2)$ in terms of $\mathbb{P}(0 < X < 2)$.
- Estimate

$$\mathbb{P}(0 < X < 2) = \int_0^2 \frac{1}{\pi(1+x^2)} dx$$

using Monte Carlo integration. Note that you can convert this into an expectation by choosing a suitable density on $[0, 2]$ (the simplest one works but be careful about constants).

- Determine your test function and compute the variance of this estimate as we have done in the class (using true value of this integral – can you deduce it since we know $\mathbb{P}(X > 2) = 0.1476$?)
- The variance of this estimate (of λ) would be the same as the variance of the main estimate I , why?

3.3 IMPORTANCE SAMPLING

While the estimators constructed using samples exactly coming from p has desirable properties as we have seen above, in the majority of cases, we need to employ more complex sampling strategies. A few cases where we need this are summarised below.

- A typical problem arises when computing tail probabilities (also called *rare events*). We may have access to samples directly from $p_*(x)$, however, sampling from the tail of $p_*(x)$ might be extremely difficult. For example, consider the Gaussian random variable X with mean 0 and variance 1. The probability of X being larger than 4 is very small, i.e., $\mathbb{P}(X > 4) \approx 0.00003$. Sampling from the tail of this density directly would be very inefficient without further tricks.
- Another typical scenario where we may want to compute expectations with respect to $p(x)$ when we do not have direct samples from it. The standard example for this is the Bayesian setting. Given a prior $p(x)$ and a likelihood $p(y|x)$, we may want to compute the expectations w.r.t. the posterior density $p(x|y)$, i.e., $\mathbb{E}_{p(x|y)}[\varphi(X)]$. In this case, we do not have access to samples from $p(x|y)$ so we need to employ other strategies.

A strategy we will pursue in this section is specific to *Monte Carlo integration*. In other words, we will next describe a strategy where we can compute integrals and expectations w.r.t. a probability density without having access to samples from it. This is slightly different than directly aiming at *sampling* from the density (which can also be used to estimate integrals). While we will look at sampling methods in the following chapters, it is important to note that importance sampling is primarily an integration technique.

3.3.1 BASIC IMPORTANCE SAMPLING

Consider the basic task of estimating the integral

$$\bar{\varphi} = \mathbb{E}_{p_*}[\varphi(X)] = \int \varphi(x)p_*(x)dx.$$

In this section, as opposed to previous sections, we assume that we cannot sample from p_* directly (or exactly, e.g., using rejection sampling³). However, we assume (in this section) we can evaluate the density $p_*(x)$ pointwise. We can still estimate this expectation (and compute integrals more generally), using samples from an instrumental, *proposal* distribution q . In other words, we can sample from a *proposal* and we can repurpose these samples to estimate expectations w.r.t. $p_*(x)$. This resembles the rejection sampling where we have also used a proposal to accept-reject samples. However, in this case, we will employ a different strategy of *weighting* samples and will not throw any of the samples away. The weights we will compute will *weight samples* so that the integral estimate gets closer to the true integral. In order to see how to do this, we compute

$$\begin{aligned} \bar{\varphi} &= \int \varphi(x)p_*(x)dx, \\ &= \int \varphi(x)\frac{p_*(x)}{q(x)}q(x)dx, && \text{“identity trick”} \end{aligned} \tag{3.13}$$

$$= \int \varphi(x)w(x)q(x)dx, \tag{3.14}$$

³Recall that rejection sampling draws i.i.d samples from the density, not *approximate*.

where $w(x) = p_*(x)/q(x)$ (which is called the *weight function*). We know from Section 3.1 that we can estimate the integral in (3.14) using samples from q . Let $X_i \sim q$ be i.i.d samples from q for $i = 1, \dots, N$. We can then estimate the integral in (3.14), hence the expectation $\bar{\varphi}$ using

$$\begin{aligned}\bar{\varphi} &= \int \varphi(x)w(x)q(x)dx \\ &\approx \frac{1}{N} \sum_{i=1}^N w_i \varphi(X_i) = \hat{\varphi}_{\text{IS}}^N,\end{aligned}\tag{3.15}$$

where $w_i = w(X_i) = p_*(X_i)/q(X_i)$ are called the *weights*. The weights will play a crucial role throughout this section. The key idea of importance sampling is that, instead of throwing away the samples by rejection, we could reweight them according to their importance. This is why this strategy is called *importance sampling* (IS).

The importance sampling algorithm for this case then can be described relatively straightforwardly. Given $p_*(x)$ (which we assume to be known in this section), we choose a proposal $q(x)$. Then, we sample $X_i \sim q(x)$ for $i = 1, \dots, N$ and compute the IS estimator as

$$\hat{\varphi}_{\text{IS}}^N = \frac{1}{N} \sum_{i=1}^N w_i \varphi(X_i),$$

where $w_i = \frac{p_*(X_i)}{q(X_i)}$ for $i = 1, \dots, N$ are the importance weights. We summarise the method in Algorithm 7. In what follows, we will discuss some details of the method.

Algorithm 7 Pseudocode for basic importance sampling

- 1: Input: The number of samples N
 - 2: **for** $i = 1, \dots, N$ **do**
 - 3: Sample $X_i \sim q(x)$
 - 4: Compute weights $w_i = \frac{p_*(X_i)}{q(X_i)}$
 - 5: **end for**
 - 6: Report the estimator
-

$$\hat{\varphi}_{\text{IS}}^N = \frac{1}{N} \sum_{i=1}^N w_i \varphi(X_i).$$

Remark 3.4. Note that one can also consider IS approximations as the approximation to the probability measure $p_*(dx)$ as

$$p_{\text{IS}}^N(dx) = \frac{1}{N} \sum_{i=1}^N w_i \delta_{X_i}(dx),$$

and hence $\hat{\varphi}_{\text{IS}}^N = (\varphi, p_{\text{IS}}^N)$. This is a useful way to think about the IS estimator later on in some proofs.

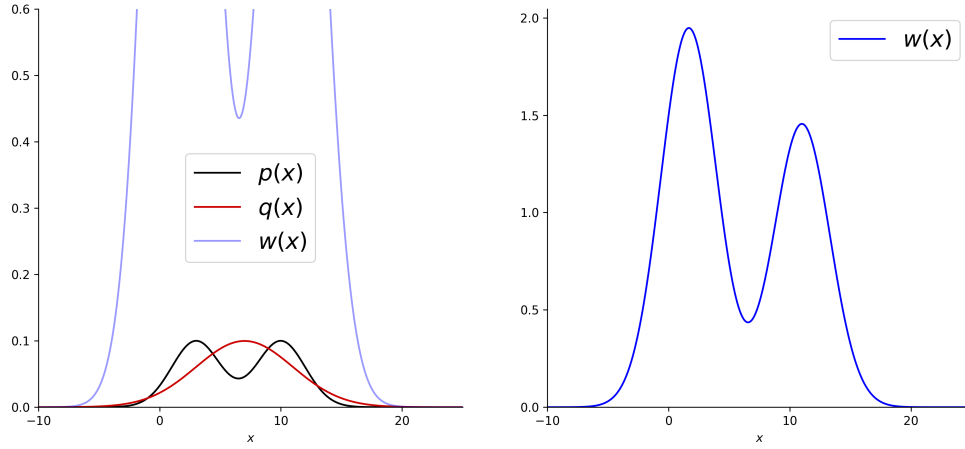


Figure 3.7: An example of target density $p_*(x)$, the proposal $q(x)$ and the associated weight function $w(x)$. One can see that if $q(x) < p_*(x)$ (which means fewer samples would be drawn from $q(x)$ in this region), then $w(x) > 1$ to account for this effect. The opposite is also true, since if $q(x) > p_*(x)$, this means that we would draw more samples than necessary, which should be downweighted, hence $w(x) < 1$ in these regions.

Remark 3.5. Unlike rejection sampling, in importance sampling, the proposal does not have to dominate the target density. Instead, the crucial requirement for the IS is that the support of the proposal should be the same as the support of the density. More precisely, we need $q(x) > 0$ whenever $p_*(x) > 0$. This is far less restrictive than the requirement of rejection sampling. Of course, the choice of proposal can still effect the performance of the IS. We will discuss this in more detail.

From Fig. 3.7, one can see an example plot of the target density $p_*(x)$, the proposal $q(x)$ and the associated weight function $w(x)$. See the caption for more details and intuition.

Example 3.6. Consider the problem of estimating $\mathbb{P}(X > 4)$ for $X \sim \mathcal{N}(0, 1)$. Describe a potential problem of using the naive MC method.

Solution. While we can exactly sample from this density, given that

$$\mathbb{P}(X > 4) = 3.16 \times 10^{-5},$$

it will be the case that very few of the samples from exact distribution will fall into this tail (Note that, while we know the exact value in this case, we will not know this in general – this is just a demonstrative example). In fact, a standard run with $N = 10000$ gives exactly zero samples that satisfy $X_i > 4$, hence provides the estimate as zero! It is obvious that this is not a great way to estimate the probability and we can use importance sampling for this. Consider a proposal $q(x) = \mathcal{N}(6, 1)$. This will draw a lot of samples from the region $X > 4$ and we can reweight this

samples w.r.t. the target density using the IS estimator in (3.15). A standard run in this case with $N = 10000$ results in

$$\hat{\varphi}_{\text{IS}}^N = 3.18 \times 10^{-5},$$

which is obviously a much closer number to the truth.

One can next prove that the estimator $\hat{\varphi}_{\text{IS}}^N$ is unbiased.

Proposition 3.3. *The estimator $\hat{\varphi}_{\text{IS}}^N$ is unbiased, i.e.,*

$$\mathbb{E}_q[\hat{\varphi}_{\text{IS}}^N] = \bar{\varphi}.$$

Proof. We simply write

$$\begin{aligned} \mathbb{E}_q[\hat{\varphi}_{\text{IS}}^N] &= \mathbb{E}_{q(x)} \left[\frac{1}{N} \sum_{i=1}^N \mathbf{w}_i \varphi(X_i) \right] \\ &= \mathbb{E}_q \left[\frac{1}{N} \sum_{i=1}^N \frac{p_*(X_i)}{q(X_i)} \varphi(X_i) \right] \\ &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_q \left[\frac{p_*(X_i)}{q(X_i)} \varphi(X_i) \right] \\ &= \frac{1}{N} \sum_{i=1}^N \int \frac{p_*(x')}{q(x')} \varphi(x') q(x') dx' \quad \text{since } X_i \sim q(x) \\ &= \int \varphi(x') p_*(x') dx', \\ &= \bar{\varphi}, \end{aligned}$$

which completes the proof. $\square$

An important quantity in IS is the *variance* of the estimator $\hat{\varphi}_{\text{IS}}^N$. The variance of the estimator is a measure of how much the estimator fluctuates around its expected value. The variance of the IS estimator (3.15) is given by the following proposition.

Proposition 3.4. *The variance of the estimator $\hat{\varphi}_{\text{IS}}^N$ is given by*

$$\text{var}_q[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \left(\mathbb{E}_q[w^2(X) \varphi^2(X)] - \bar{\varphi}^2 \right).$$

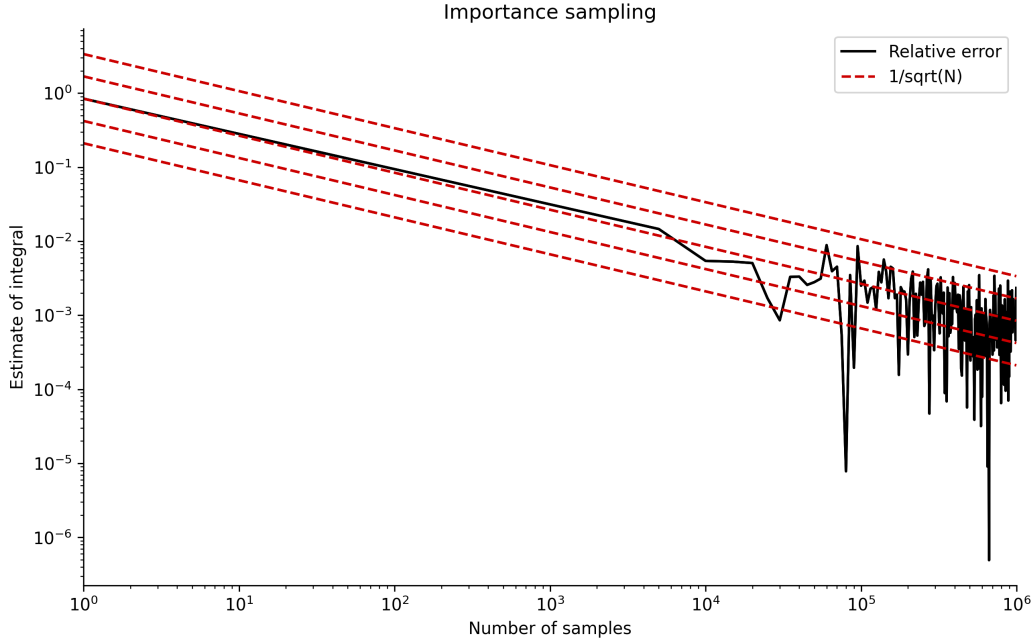


Figure 3.8: The importance sampling estimator $\hat{\varphi}_{\text{IS}}^N$ is plotted against the number of samples N for the example in in Fig . 3.7, for $\varphi(x) = x^2$. This demonstrates that the random error in the IS case also satisfies $\mathcal{O}(1/\sqrt{N})$ convergence rate.

Proof. Next we write out the estimator $\hat{\varphi}_{\text{IS}}^N$ in (3.15)

$$\begin{aligned}
 \text{var}_q[\hat{\varphi}_{\text{IS}}^N] &= \text{var}_q \left[\frac{1}{N} \sum_{i=1}^N w_i \varphi(X_i) \right] \\
 &= \frac{1}{N^2} \text{var}_q \left[\sum_{i=1}^N w(X_i) \varphi(X_i) \right] \\
 &= \frac{1}{N} \text{var}_q [w(X) \varphi(X)] \quad \text{where } X \sim q(x) \\
 &= \frac{1}{N} \left(\mathbb{E}_q [w^2(X) \varphi^2(X)] - \mathbb{E}_q [w(X) \varphi(X)]^2 \right) \\
 &= \frac{1}{N} \left(\mathbb{E}_q [w^2(X) \varphi^2(X)] - \bar{\varphi}^2 \right),
 \end{aligned}$$

which concludes the proof. We have used the fact that the variance of the sum of independent random variables is the sum of the variances. $\square$

One can see that this easily leads to the bound for the standard deviation $\text{std}_q[\hat{\varphi}_{\text{IS}}^N] \leq \mathcal{O} \left(\frac{1}{\sqrt{N}} \right)$. Also, we still have the result for the relative absolute error as

$$|\hat{\varphi}_{\text{IS}}^N - \bar{\varphi}| \leq \frac{V}{\sqrt{N}},$$

where V is an almost surely finite random variable. As in the perfect MC case, we will not prove this result as it is beyond our scope, but curious reader can refer to Corollary 2.2 in [Akyildiz \(2019\)](#)

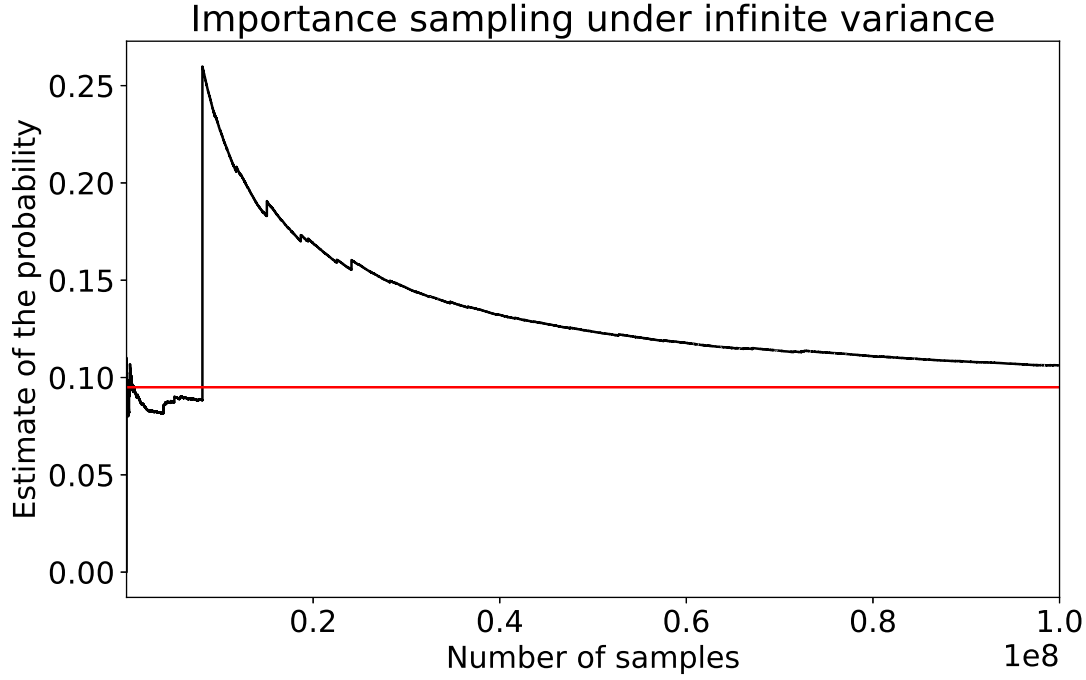


Figure 3.9: Estimating $\mathbb{P}(2 < X < 6)$ where X is Cauchy with $q(x) = \mathcal{N}(0, 1)$. The true value is plotted in red and the estimator value in black.

(which also holds for the self normalised case which will be introduced below). A demonstration of this rate for importance sampling can be seen from Fig. 3.8.

We can see that the variance of the IS estimator is finite if

$$\mathbb{E}_q[w^2(X)\varphi^2(X)] < \infty.$$

This implies that

$$\int w^2(x)\varphi^2(x)q(x)dx = \int \frac{p_\star^2(x)}{q(x)}\varphi^2(x)p_\star(x)dx < \infty.$$

In other words, for our importance sampling estimate to be well-defined, the ratio

$$\frac{p_\star^2(x)}{q(x)}\varphi^2(x)$$

has to be integrable. We will see next an example where this condition is not satisfied.

Example 3.7 (Infinite variance IS, Example 3.8 from Robert and Casella (2010)). Consider the target

$$p_\star(x) = \frac{1}{\pi} \frac{1}{1+x^2},$$

which is the Cauchy density. Let us choose the proposal

$$q(x) = \mathcal{N}(x; 0, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

Discuss whether the IS estimator is well-defined.

Solution. We can compute the ratio $\frac{p_*(x)}{q(x)} \propto \exp(x^2/2)/(1+x^2)$ and see that it is explosive (precisely, its integral is infinite). This can result in problematic situations even if φ ensures that the variance is finite. For example, consider the problem of estimating $\mathbb{P}(2 < X < 6)$. One example run for this case can be seen from Fig. 3.9. One can see that the estimator in this case is unstable and cannot be reliably used.

Remark 3.6 (Optimal proposal). We can try to inspect the variance expression to figure out which proposals can give us variance reduction. From Prop. 3.4, it follows that we have

$$\text{var}_q[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \text{var}_q[w(X)\varphi(X)].$$

This means that minimising the variance of the IS estimator is the same as minimising the variance of the function $w(x)\varphi(x)$. Moreover, looking at the expression,

$$\text{var}_q[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \left(\mathbb{E}_q[w^2(X)\varphi^2(X)] - \bar{\varphi}^2 \right),$$

we can see that since $\bar{\varphi}^2 > 0$ (which is independent of the proposal), we should choose a proposal that minimises $\mathbb{E}_q[w^2(X)\varphi^2(X)]$. We can lower bound this quantity using Jensen's inequality:

$$\mathbb{E}_q[w^2(X)\varphi^2(X)] \geq \mathbb{E}_q[|w(X)\varphi(X)|^2],$$

where we used the fact that $(\cdot)^2$ is a convex function (For a convex function f , Jensen's inequality states that $\mathbb{E}_q[f(X)] \geq f(\mathbb{E}_q[X])$). Using $w(x) = p_*(x)/q(x)$, we arrive at the following lower bound:

$$\mathbb{E}_q[w^2(X)\varphi^2(X)] \geq \mathbb{E}_{p_*}[|\varphi(X)|^2]. \quad (3.16)$$

Now let us expand the term $\mathbb{E}_q[w^2(X)\varphi^2(X)]$ out and write

$$\begin{aligned} \mathbb{E}_q[w^2(X)\varphi^2(X)] &= \mathbb{E}_q\left[\frac{p_*^2(X)}{q^2(X)}\varphi^2(X)\right] \\ &= \int \frac{p_*^2(x)}{q^2(x)}\varphi^2(x)q(x)dx \\ &= \int p_*(x)\frac{p_*(x)}{q(x)}\varphi^2(x)dx, \\ &= \mathbb{E}_{p_*}[w(X)\varphi^2(X)]. \end{aligned} \quad (3.17)$$

The last equation, eq. (3.17), suggests that we can choose a proposal such that we attain the lower bound of this function (3.16) (which means that it would be the minimiser). In particular, if we choose a proposal $q(x)$ such that

$$w(x) = \frac{p_*(x)}{q(x)} = \frac{\mathbb{E}_{p_*}[|\varphi(X)|]}{|\varphi(x)|}$$

is satisfied, then (3.17) would be equal to the lower bound (3.16). This implies that

$$q_*(x) = p_*(x) \frac{|\varphi(x)|}{\mathbb{E}_{p_*}[|\varphi(X)|]}, \quad (3.18)$$

would minimise the variance of the importance sampling estimator.

Choosing q_* as the proposal, one can see that the variance of the IS estimator satisfies

$$\begin{aligned} \text{var}_{q_*}[\hat{\varphi}_{\text{IS}}^N] &= \frac{1}{N} \mathbb{E}_{p_*}[|\varphi(X)|]^2 - \frac{1}{N} \bar{\varphi}^2 \\ &\leq \frac{1}{N} \mathbb{E}_{p_*}[\varphi^2(X)] - \frac{1}{N} \bar{\varphi}^2 \\ &= \text{var}_{p_*}[\hat{\varphi}_{\text{MC}}^N], \end{aligned}$$

therefore we obtain

$$\text{var}_{q_*}[\hat{\varphi}_{\text{IS}}^N] \leq \text{var}_{p_*}[\hat{\varphi}_{\text{MC}}^N],$$

i.e., a variance reduction. In fact, one can show that, if $\varphi(x) \geq 0$ for all $x \in \mathbb{R}$, then the variance of the IS estimator with optimal proposal q_* is equal to zero.

We note that this optimal construction of the proposal (3.18) is not possible to implement in practice. It requires the knowledge of the very quantity we want to estimate, namely, $\mathbb{E}_{p_*}[|\varphi(X)|]$! But in general, we can choose proposals that minimise the variance of the IS estimator where possible. This idea has been used in the literature to construct proposals that minimise the variance of the estimator, see, e.g., [Akyildiz and Míguez \(2021\)](#) and references therein. We will construct some simple examples next.

3.3.2 CHOOSING THE OPTIMAL IMPORTANCE SAMPLING PROPOSAL WITHIN A FAMILY

While we may not be able to choose the optimal proposal q_* as shown above (as it requires us to know the target density), we can still optimise the proposal within a family of distributions. Recall the discussion for this in the case of rejection sampling, where we optimised the acceptance rate $\hat{a}$ in order to choose the optimal parameter of a proposal. In the case of importance sampling, the target is to *minimise* the variance. Let q_θ be a parametric family of proposals. Recall that in Prop. 3.4, we have defined the variance of the IS estimator as

$$\text{var}_{q_\theta}[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \left(\mathbb{E}_{q_\theta} [w_\theta^2(X) \varphi^2(X)] - \bar{\varphi}^2 \right),$$

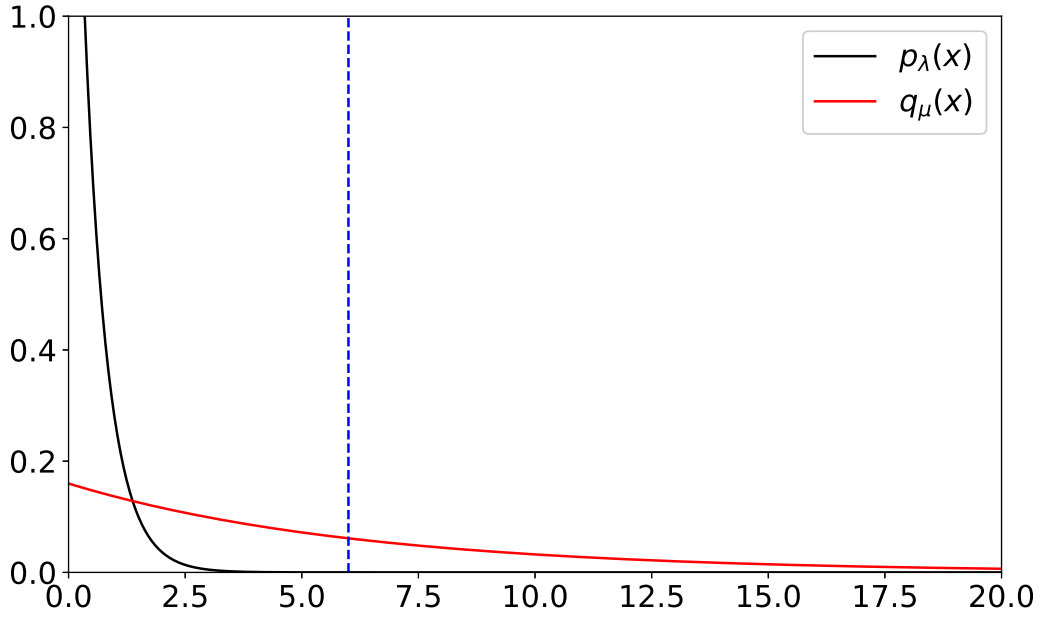


Figure 3.10: The density of the exponential distribution p_λ , the proposal q_μ and $K = 6$

where $\bar{\varphi} = \mathbb{E}_{p_\star}[\varphi(X)]$ and

$$w_\theta(x) = \frac{p_\star(x)}{q_\theta(x)}.$$

Note that $1/N$ does not change the location of the minimum and $\bar{\varphi}^2$ is independent of θ . Therefore, we drop these terms and see that in order to minimise this variance w.r.t. θ , we need to solve the following optimisation problem

$$\theta_\star = \operatorname{argmin}_\theta \mathbb{E}_{q_\theta} [w_\theta^2(X) \varphi^2(X)].$$

We will now demonstrate this with an example.

Example 3.8 (Minimum variance IS). We are given an exponential distribution

$$p_\lambda(x) = \lambda \exp(-\lambda x),$$

and want to compute $\mathbb{P}(X > K)$. For example, $\lambda = 2$ and $K = 6$, we can analytically compute $\mathbb{P}(X > 6) = 6.144 \times 10^{-6}$. Therefore, the standard MC estimator would be very inefficient to compute this probability. In order to mitigate the problem, we would like to use another exponential proposal $q_\mu(x)$ which may have higher probability concentration around 6. We would like to design our proposal using the minimum variance criterion (see Remark 3.6). Find the optimal $\mu_\star$ that would make the estimator variance minimal.

Solution. Accordingly, we would like to find μ such that

$$\mu_\star = \operatorname{argmin}_{\mu} \mathbb{E}_q \left[w^2(X) \varphi^2(X) \right].$$

In this case, note that we have $\varphi(x) = \mathbf{1}_{\{x > K\}}(x)$. In order to do this, we write next

$$\begin{aligned} \mathbb{E}_{q_\mu} \left[\left(w^2(X) \varphi^2(X) \right) \right] &= \int \frac{p_\lambda(x)^2}{q_\mu(x)^2} q_\mu(x) \varphi^2(x) dx, \\ &= \int_K^\infty \frac{p_\lambda(x)}{q_\mu(x)} p_\lambda(x) dx, \\ &= \int_K^\infty \frac{\lambda^2 e^{-2\lambda x}}{\mu e^{-\mu x}} dx, \\ &= \frac{\lambda^2}{\mu} \int_K^\infty e^{-(2\lambda - \mu)x} dx. \end{aligned}$$

Note at this stage that in order for this integral to be finite, we need to have $2\lambda - \mu > 0$. Therefore, we limit for $\mu \in (0, 2\lambda)$. In order to compute this, we can multiply and divide by $(2\lambda - \mu)$ and obtain

$$\mathbb{E}_{q_\mu} \left[\left(w^2(X) \varphi^2(X) \right) \right] = \frac{\lambda^2}{\mu(2\lambda - \mu)} \int_K^\infty (2\lambda - \mu) e^{-(2\lambda - \mu)x} dx,$$

and using the CDF of the exponential distribution, we obtain

$$g(\mu) = \mathbb{E}_{q_\mu} \left[\left(w^2(X) \varphi^2(X) \right) \right] = \frac{\lambda^2}{\mu(2\lambda - \mu)} \left[1 - 1 + e^{-(2\lambda - \mu)K} \right]. \quad (3.19)$$

Now we optimise $g(\mu)$ w.r.t. μ . As usual, we compute first log (and drop the terms unrelated to μ as they will not matter in optimisation)

$$\log g(\mu) =^c -\log \mu - \log(2\lambda - \mu) + \mu K.$$

Computing

$$\frac{d}{d\mu} \log g(\mu) = -\frac{1}{\mu} + \frac{1}{2\lambda - \mu} + K,$$

Setting this to zero, we obtain

$$\begin{aligned} -\frac{1}{\mu} + \frac{1}{2\lambda - \mu} + K &= 0, \\ \Rightarrow -(2\lambda - \mu) + \mu + K\mu(2\lambda - \mu) &= 0, \\ \Rightarrow K\mu^2 - 2K\mu\lambda + 2\lambda - 2\mu &= 0, \\ \Rightarrow K\mu^2 - 2(K\lambda + 1)\mu + 2\lambda &= 0. \end{aligned}$$

This is a quadratic equation, therefore we will have two solutions:

$$\begin{aligned}\mu &= \frac{2(K\lambda + 1) \pm \sqrt{(2K\lambda + 2)^2 - 8K\lambda}}{2K}, \\ &= \frac{2(K\lambda + 1) \pm \sqrt{4K^2\lambda^2 + 4}}{2K}.\end{aligned}$$

If we inspect this solution, if we choose μ to be the sum of the two terms, we will then have $\mu > 2\lambda$ which is a violation of a condition we imposed for the integral to be finite. Therefore, we arrive at

$$\mu_* = \frac{2(K\lambda + 1) - \sqrt{4K^2\lambda^2 + 4}}{2K}.$$

After this tedious computation, we can now verify the reduction in variance and estimation quality. Let us now set $K = 6$ and $\lambda = 2$. See Fig. 3.10 for plot of p_λ , $K = 6$ and q_{μ_*} (the optimal exponential proposal). We can see that the proposal puts much higher mass to the right of K . A standard run for $N = 10^5$ samples gives us **zero** samples in the region of $X > 6$, therefore the standard MC estimate is zero! Compared to $\hat{\varphi}_{\text{MC}}^N = 0$, using q_{μ_*} as a proposal, we obtain $\hat{\varphi}_{\text{IS}}^N = 6.08 \times 10^{-6}$ which is a much better estimate.

Let us compare the theoretical variances of two estimators. The standard variance of $\hat{\varphi}_{\text{MC}}^N$ is

$$\text{var}_p(\hat{\varphi}_{\text{MC}}^N) = \frac{1}{N} \text{var}_p(\varphi(X)),$$

where

$$\begin{aligned}\text{var}_p(\varphi(X)) &= \int \varphi(x)^2 p_\lambda(x) dx - \left(\int \varphi(x) p_\lambda(x) dx \right)^2, \\ &= \int_K^\infty p_\lambda(x) dx - \left(\int_K^\infty p_\lambda(x) dx \right)^2.\end{aligned}$$

Using CDFs, we can compute this quantity hence can obtain the estimate of the variance for $\hat{\varphi}_{\text{MC}}^N$.

Now set $\mu = \mu_*$. The variance of $\hat{\varphi}_{\text{IS}}^N$ is given by (see Prop. 3.4)

$$\text{var}_q[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \left(\mathbb{E}_q[w^2(X)\varphi^2(X)] - \bar{\varphi}^2 \right),$$

We have already computed the term $\mathbb{E}_q[w^2(X)\varphi^2(X)]$ in Eq. (3.19). The second term is the true integral, which we also summarised how to compute above, i.e., $\bar{\varphi} = \int_K^\infty p_\lambda(x) dx$ which can be computed using the exponential CDF. In this particular case, we compute

$$\text{var}_q[\hat{\varphi}_{\text{IS}}^N] = \frac{1}{N} \left(g(\mu_*) - \bar{\varphi}^2 \right).$$

The theoretical variance of the naive MC estimator is given by 6.14×10^{-7} for $N = 10$ samples vs. the IS estimator variance is 6.04×10^{-11} for the same amount of samples. This is a huge improvement in the variance of the estimation.

Exercise 3.6. Minimum variance is a good design principle when choosing proposals. However, as can be seen from Example 3.8, this can be tedious. In this example, we will artificially simplify this process in order to not solve quadratic equations to find optimum. Let $p(x)$ be the exponential distribution with $\lambda = 1$, i.e.,

$$p(x) = e^{-x}, \quad x \geq 0.$$

Assume that we would like to estimate the expectation of $\varphi(x)$ w.r.t. this distribution where

$$\varphi(x) = e^{-\frac{Kx}{2}}$$

where $K > 0$. While this test function is artificial, it is good for testing your skills. Consider the exponential proposal

$$q_\mu(x) = \mu e^{-\mu x}.$$

Find the optimal $\mu_\star$ in terms of K that minimises the variance of the IS estimate (see Example 4.9 in the lecture notes – but the computation here is much simpler!).

Once you obtain $\mu_\star$ in terms of K , then evaluate $\text{var}_{q_{\mu_\star}}(\hat{\varphi}_{\text{IS}}^N)$ and $\text{var}_p(\bar{\varphi}_{\text{MC}}^N)$ (MC is the case of the usual Monte Carlo estimator). Note that both cases do not require any sampling (find them analytically). Verify the variance reduction.

Exercise 3.7. Let $p(x) = \mathcal{N}(x; 0, 1)$ and $\varphi(x) = x$.

- Verify that the standard MC estimator, $X_1, \dots, X_N \sim p(x)$:

$$\hat{\varphi}_{\text{MC}}^N = \frac{1}{N} \sum_{i=1}^N \varphi(X_i) = \frac{1}{N} \sum_{i=1}^N X_i,$$

has variance $1/N$, i.e.,

$$\text{var}_p(\hat{\varphi}_{\text{MC}}^N) = \frac{1}{N}.$$

What would be the variance if we had $p(x) = \mathcal{N}(x; \mu, \sigma^2)$?

- Let

$$q_\lambda(x) = \mathcal{N}(x; 0, 1/\lambda).$$

Find the minimum variance IS proposal, i.e., find $\lambda_\star$ that minimises the IS estimator variance (again follow Example 4.9 and no quadratic equations are needed for this case either). Show that the variance is less than $1/N$ at this value by computing the exact number.

- Implement the MC sampler and IS sampler using this optimal λ_* . Compute the estimate of the mean ($= 0$) and verify that your estimators work. Then compute the variance of your estimator empirically (in other words, choose N and run the same experiment M times (Monte Carlo runs)). Verify that the IS estimator variance is, indeed, less than naive MC estimator variance!

Exercise 3.8 (Stochastic Simulation, Final Exam Q2, 2024). Consider the following Rayleigh density function:

$$p(x) = x \exp\left(-\frac{x^2}{2}\right), \quad x \in [0, \infty).$$

Suppose we would like to estimate the **mean** of this distribution.

- (a) Consider the importance sampling (IS) task for this problem. We would like to use another Rayleigh density function $q_\lambda(x)$ as the proposal:

$$q_\lambda(x) = \frac{x}{\lambda^2} \exp\left(-\frac{x^2}{2\lambda^2}\right), \quad x \in [0, \infty), \quad \lambda > 0.$$

- (i) State the test function $\varphi(x)$ for this problem. Show that $\int \varphi(x)p(x)dx = \sqrt{\pi/2}$.

Hint: Gaussian second moment identity: $\int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx = 1$.

- (ii) State the IS estimator $\hat{\varphi}_{\text{IS}}^N$ in terms of N samples $X_1, \dots, X_N \sim q_\lambda(x)$.
 (iii) State the variance expression $\text{var}(\hat{\varphi}_{\text{IS}}^N)$ in terms of p, φ, q_λ , and N (You do not have to derive it).
 (iv) By explicitly computing the quantity you provided in Part (a)(iii), show that

$$\text{var}(\hat{\varphi}_{\text{IS}}^N) = \frac{1}{N} \left(2\lambda^6(2\lambda^2 - 1)^{-2} - \frac{\pi}{2} \right).$$

Hint: You can use integration by parts (or any other technique of your choice). In integration by parts, recognise that the second term can be completed to an integral of a Rayleigh density.

- (v) Show that the optimal λ_* that minimises the variance is $\lambda_* = \sqrt{3/2}$.
 (vi) Derive the expression for the variance $\text{var}(\hat{\varphi}_{\text{IS}}^N)$ with this optimal choice λ_* .
 (b) Now consider perfect Monte Carlo (MC) sampling for the same problem with the same test function used in Part (a).
 (i) Provide the MC estimator $\hat{\varphi}_{\text{MC}}^N$ in terms of N samples $X_1, \dots, X_N \sim p(x)$.
 (ii) State the variance expression $\text{var}(\hat{\varphi}_{\text{MC}}^N)$ in terms of p, φ , and N (You do not have to derive it).

(iii) By explicitly computing the quantity you provided in Part (b)(ii), show that

$$\text{var}(\hat{\varphi}_{\text{MC}}^N) = \frac{1}{N} \left(2 - \frac{\pi}{2} \right).$$

Hint: The computation here is a special case of Part (a)(iv), so you can directly compute the integral using your earlier result.

(iv) Compare $\text{var}(\hat{\varphi}_{\text{MC}}^N)$ and $\text{var}(\hat{\varphi}_{\text{IS}}^N)$ derived in Part (a)(vi) and discuss why we would perform IS rather than MC even if we have access to i.i.d samples.

Exercise 3.9. Let us consider the standard Gaussian model

$$\begin{aligned} p(x) &= \mathcal{N}(x; 0, 1) \\ p(y|x) &= \mathcal{N}(y; x, 1), \end{aligned}$$

and, again, we would like to estimate the marginal likelihood $p(y) = \int p(y|x)p(x)dx$ using i.i.d Monte Carlo (MC) and importance sampling (IS). This is a quantity that computes the model's likelihood under a particular observation.

1. Write the analytical expression of $p(y)$ (this is provided in lecture notes – you don't have to prove it). Now assume that we observe $y = 9$. Compute $p(y = 9)$ using your exact $p(y)$ expression. (Hint: It is a small number)
2. What happened? We observed an unlikely observation that could come from the model. It is of practical interest to estimate such probabilities (unlikely events under the model). Next, implement the standard MC estimator (i.i.d) to estimate the integral $p(y = 9) = \int p(y = 9|x)p(x)dx$ by drawing i.i.d samples from $p(x)$. Describe your test function and estimator clearly. Provide results for $N = 10, 100, 1000, 10000, 100000$ by computing the relative absolute error (RAE) (note that you know the true value from Part 1, therefore use that to compute your RAE) for these N and plot it w.r.t. N .
3. Next choose a proposal of the form

$$q(x) = \mathcal{N}\left(x; \mu, \frac{1}{2}\right).$$

For any observation y , find the minimum variance proposal mean, i.e., μ_* that minimises the variance in terms of y . Then set $y = 9$ and compute μ_* for this case. Implement the IS estimator with μ_* as the proposal mean and compute the RAE for $N = 10, 100, 1000, 10000, 100000$ by computing the relative absolute error (RAE) (note that you know the true value from Part 1, therefore use that to compute your RAE) for these N and plot it w.r.t. N .

4. Compare the RAE for MC and IS to each other (provide a plot w.r.t. N by plotting them together). What is the difference? Comment about the difference about the accuracy of these two estimators.

Use `plt.loglog` to plot throughout this exercise to have clear figures.

3.3.3 SELF-NORMALISED IMPORTANCE SAMPLING

As mentioned several times in past chapters, in many scenarios, we have access to the *unnormalised* density, i.e., given p_* , we can evaluate it up to a normalising constant. As usual, we denote this density $\bar{p}_*(x)$ and recall that it is related to p_* by

$$p_*(x) = \frac{\bar{p}_*(x)}{Z}$$

where $Z = \int \bar{p}_*(x)dx$. In the context of Bayesian inference, we usually have an unnormalised posterior density $\bar{p}_*(x) := \bar{p}(x|y) \propto p(y|x)p(x)$. In the previous section, we have built an importance sampling estimator for the case where we have access to the normalised density. In this section, we will generalise the idea and assume we only have access to the unnormalised density.

Consider, again, the problem of estimating expectations of a given density p_* . For the case where we can only evaluate $\bar{p}_*(x)$, one way to estimate this expectation is to sample from a proposal distribution q and rewrite the integral as

$$\begin{aligned} \bar{\varphi} &= \int \varphi(x)p_*(x)dx, \\ &= \frac{\int \varphi(x) \frac{\bar{p}_*(x)}{q(x)} q(x)dx}{\int \frac{\bar{p}_*(x)}{q(x)} q(x)dx}, \end{aligned} \quad (3.20)$$

where we use the fact that $p_*(x) = \bar{p}_*(x)/Z$. This gives us two separate integration problems, one to estimate the numerator and one to estimate the denominator. We will estimate both quantities using samples from $q(x)$.

Let us now introduce the unnormalised weight function $W(x)$

$$W(x) = \frac{\bar{p}_*(x)}{q(x)},$$

which is analogous to the normalised weight function $w(x)$ in the previous section. Using $X_i \sim q(x)$ and building the Monte Carlo estimator of the numerator and denominator, we arrive at the following estimator of the (3.20):

$$\begin{aligned} \hat{\varphi}_{\text{SNIS}}^N &= \frac{\frac{1}{N} \sum_{i=1}^N \varphi(X_i)W(X_i)}{\frac{1}{N} \sum_{i=1}^N W(X_i)}, \\ &= \frac{\sum_{i=1}^N \varphi(X_i)W(X_i)}{\sum_{i=1}^N W(X_i)} \\ &= \sum_{i=1}^N \bar{w}_i \varphi(X_i), \end{aligned} \quad (3.21)$$

where

$$\bar{w}_i = \frac{W(X_i)}{\sum_{i=1}^N W(X_i)},$$

Algorithm 8 Pseudocode for self-normalised importance sampling

- 1: Input: The number of samples N
- 2: **for** $i = 1, \dots, N$ **do**
- 3: Sample $X_i \sim q(x)$
- 4: Compute unnormalised weights $W(X_i) = \frac{\bar{p}_*(X_i)}{q(X_i)}$
- 5: Normalise:

$$\bar{w}_i = \frac{W(X_i)}{\sum_{i=1}^N W(X_i)}.$$

- 6: **end for**
- 7: Report the estimator

$$\hat{\varphi}_{\text{SNIS}}^N = \sum_{i=1}^N \bar{w}_i \varphi(X_i).$$

are the *normalised* weights. This estimator (3.21) is called the self-normalised importance sampling (SNIS) estimator. The full scheme given in Algorithm 8.

Remark 3.7 (Bias and variance). As opposed to the normalised case, the estimator $\hat{\varphi}_{\text{SNIS}}^N$ is *biased*. The reason of this bias can be seen by recalling the integral (3.20). By sampling from $q(x)$, we can construct unbiased estimates of the numerator and denominator. However, the ratio of these two quantities is biased in general. However, it can be shown that the bias of the SNIS estimator decreases with a rate $\mathcal{O}(1/N)$ (Agapiou et al., 2017).

Since the SNIS estimator is biased, we cannot use the same variance formula as in the previous section. It makes sense to consider the $\text{MSE}(\hat{\varphi}_{\text{SNIS}}^N)$ instead. However, this quantity is challenging to control in general – without bounded test functions. With bounded test functions, it is possible to show that the $\text{MSE}(\hat{\varphi}_{\text{SNIS}}^N)$ is controlled with a rate $\mathcal{O}(1/N)$ (Agapiou et al., 2017; Akyildiz and Míguez, 2021), which we show below.

In the following, we provide the MSE bound for the SNIS estimator see, e.g., Agapiou et al. (2017) or Akyildiz and Míguez (2021). We use here the notation of Akyildiz and Míguez (2021, Theorem 1) for the proof.

Proposition 3.5. Assume for simplicity that $\sup_x |\varphi(x)| \leq 1$. Then,

$$\text{MSE}(\hat{\varphi}_{\text{SNIS}}^N) = \mathbb{E}_q \left[(\hat{\varphi}_{\text{SNIS}}^N - \bar{\varphi})^2 \right] \leq \frac{4}{N} \frac{\mathbb{E}_q [W^2(X)]}{(\mathbb{E}_q [W(X)])^2}.$$

Proof. We start by recalling our notation for integrals $(\varphi, p_*) := \int \varphi(x) p_*(x) dx$ and Monte Carlo approximations $(\varphi, p_*^N) := (1/N) \sum_{i=1}^N \varphi(X_i)$ for $X_i \sim p_*$ which will come in handy in

this proof. Let us write our SNIS approximation of the measure $p_\star$ as (recall Remark 3.4)

$$p_{\text{SNIS}}^N(dx) = \frac{1}{N} \sum_{i=1}^N \bar{w}_i \delta_{X_i}(dx),$$

where $X_i \sim q$, hence that $\hat{\varphi}_{\text{SNIS}}^N = (\varphi, p_{\text{SNIS}}^N)$. Let us also recall in this notation

$$\bar{\varphi} = (\varphi, p_\star) = \int \varphi(x) p_\star(x) dx = \frac{\int \varphi(x) \frac{\bar{p}_\star(x)}{q(x)} q(x) dx}{\int \frac{\bar{p}_\star(x)}{q(x)} q(x) dx} = \frac{(\varphi W, q)}{(W, q)}.$$

With a similar derivation, we can also write

$$\hat{\varphi}_{\text{SNIS}}^N = \frac{(\varphi W, q^N)}{(W, q^N)}.$$

Now next we write

$$\begin{aligned} |(\varphi, p) - (\varphi, p_{\text{SNIS}}^N)| &= \left| \frac{(\varphi W, q)}{(W, q)} - \frac{(\varphi W, q^N)}{(W, q^N)} \right| \\ &= \left| \frac{(\varphi W, q)}{(W, q)} - \frac{(\varphi W, q^N)}{(W, q^N)} + \frac{(\varphi W, q^N)}{(W, q)} - \frac{(\varphi W, q^N)}{(W, q)} \right| \\ &\leq \left| \frac{(\varphi W, q)}{(W, q)} - \frac{(\varphi W, q^N)}{(W, q)} \right| + \left| \frac{(\varphi W, q^N)}{(W, q)} - \frac{(\varphi W, q^N)}{(W, q^N)} \right| \\ &= \frac{1}{(W, q)} \left| (\varphi W, q) - (\varphi W, q^N) \right| + \left| (\varphi W, q^N) \right| \left| \frac{1}{(W, q)} - \frac{1}{(W, q^N)} \right|. \end{aligned}$$

At this point, we use $\sup_x |\varphi(x)| \leq 1$ (note that, this is without loss of generality for bounded test functions) and get $|(\varphi W, q^N)| \leq |(W, q^N)|$. Using this, we can write

$$\begin{aligned} |(\varphi, p) - (\varphi, p_{\text{SNIS}}^N)| &\leq \frac{1}{(W, q)} \left| (\varphi W, q) - (\varphi W, q^N) \right| + |(W, q^N)| \left| \frac{(W, q^N) - (W, q)}{(W, q^N)(W, q)} \right| \\ &= \frac{1}{(W, q)} \left(\left| (\varphi W, q) - (\varphi W, q^N) \right| + \left| (W, q^N) - (W, q) \right| \right). \end{aligned}$$

Next, we take squares of both sides (since both sides are nonnegative), use $(a + b)^2 \leq 2(a^2 + b^2)$ and take expectations

$$\begin{aligned} \mathbb{E} \left[\left| (\varphi, p) - (\varphi, p_{\text{SNIS}}^N) \right|^2 \right] &\leq 2\mathbb{E} \left[\frac{1}{(W, q)^2} \left(\left| (\varphi W, q) - (\varphi W, q^N) \right|^2 + \left| (W, q^N) - (W, q) \right|^2 \right) \right] \\ &= \frac{2}{(W, q)^2} (\text{var}_q[\varphi W, q] + \text{var}_q[W, q]), \\ &\leq \frac{2}{(W, q)^2} (\mathbb{E}_q[\varphi^2(X) W^2(X)] + \mathbb{E}_q[W^2(X)]), \end{aligned}$$

by using $\sup_x |\varphi(x)| \leq 1$, we conclude the proof. $\square$

3.4 BAYESIAN INFERENCE WITH IMPORTANCE SAMPLING

In this section, we demonstrate the applications of the Monte Carlo and importance sampling estimators to Bayesian inference. Self normalised IS is a natural choice for Bayesian inference. Assume that we have a prior $p(x)$ and a likelihood $p(y|x)$. The posterior is given by

$$p(x|y) = \frac{p(y|x)p(x)}{\int p(y|x)p(x)dx}.$$

Let $\bar{p}(x|y) := p(y|x)p(x)$ as usual and design an importance sampler to estimate expectations of the form

$$\mathbb{E}_{p(x|y)}[\varphi(x)] = \int \varphi(x)p(x|y)dx.$$

Assume that we choose $q(x)$ and decided to perform SNIS. We first sample $X_1, \dots, X_N \sim q(x)$ and construct

$$W_i = \frac{\bar{p}(X_i|y)}{q(X_i)} = \frac{p(y|X_i)p(X_i)}{q(X_i)}.$$

We can now normalise these weights and obtain

$$\bar{w}_i = \frac{W_i}{\sum_{i=1}^N W_i},$$

which will give us the Monte Carlo estimator:

$$\mathbb{E}_{p(x|y)}[\varphi(x)] \approx \sum_{i=1}^N \bar{w}_i \varphi(X_i).$$

It is useful to recall that this estimator is biased, since it is a SNIS estimator. However, as a byproduct of this estimator, we can also obtain an **unbiased** estimate of the marginal likelihood $p(y)$. Note that, this is already provided by the SNIS estimator

$$p(y) \approx \frac{1}{N} \sum_{i=1}^N W_i.$$

Proposition 3.6. *The marginal likelihood estimator given by*

$$p^N(y) = \frac{1}{N} \sum_{i=1}^N W_i,$$

is an unbiased estimator of the marginal likelihood $p(y)$.

Proof. We can easily show this

$$\begin{aligned}
\mathbb{E}_q \left[\sum_{i=1}^N W_i \right] &= \sum_{i=1}^N \mathbb{E}_q \left[\frac{p(y|X_i)p(X_i)}{q(X_i)} \right] \\
&= N \mathbb{E}_q \left[\frac{p(y|X)p(X)}{q(X)} \right] \\
&= N \int \frac{p(y|x)p(x)}{q(x)} q(x) dx \\
&= Np(y).
\end{aligned}$$

□

As we have seen before, a number of interesting problems require computing normalising constants, including model selection and prediction. SNIS estimators are very useful in the sense that they provide an unbiased estimate of it. Let us now look at an example using the optimal importance proposal as introduced in Remark 3.6.

Example 3.9 (Marginal likelihood using optimal importance proposal). We have seen that we can get unbiased estimates of the marginal likelihood. Find the optimal proposal q_* for estimating the marginal likelihood.

Solution. We will now see how to find the optimal importance proposal to compute the marginal likelihood. Note that, we have

$$p(y) = \int p(y|x)p(x)dx,$$

for some prior $p(x)$ and likelihood $p(y|x)$. Note, as we mentioned before, in this case $p(x)$ can be seen as the distribution to sample from and $\varphi(x) = p(y|x)$ to obtain the standard problem of integration $\int \varphi(x)p(x)dx$. A naive way to approximate this quantity (as we have seen before) is to sample i.i.d from $p(x)$ and approximate the integral, i.e., $X_1, \dots, X_N \sim p(x)$ and write

$$p_{\text{MC}}^N(y) = \frac{1}{N} \sum_{i=1}^N \varphi(X_i) = \frac{1}{N} \sum_{i=1}^N p(y|X_i).$$

We can now look at the variance of this estimator

$$\text{var}_p \left[p_{\text{MC}}^N(y) \right] = \frac{1}{N} \text{var}_{p(x)}[p(y|x)].$$

This quantity may depend on the prior-likelihood selection and can be large.

Let us take Remark 3.6 seriously as the question suggested and search for the optimal proposal q_* . From (3.18), we can see that

$$q_*(x) = p(x) \frac{|\varphi(x)|}{\mathbb{E}_p(x)[|\varphi(x)|]}.$$

In this case, however, we have $\varphi(x) = p(y|x)$ (and $|\varphi(x)| = \varphi(x)$ since the likelihood is positive everywhere). We can now write

$$q_*(x) = p(x) \frac{p(y|x)}{\mathbb{E}_p[p(y|x)]} = p(x) \frac{p(y|x)}{p(y)}.$$

In other words, the optimal proposal is the posterior itself! Now we can compute the IS estimator variance where we plug $q_* = p(x|y)$. Note to explore variance, we write

$$\text{var}_{q_*}[p_{\text{IS}}^N(y)] = \frac{1}{N} \left(\mathbb{E}_{q_*} \left[\left(\frac{p(x)}{q_*(x)} \right)^2 p(y|x)^2 \right] - p(y)^2 \right).$$

We compute the first term in brackets,

$$\begin{aligned} \mathbb{E}_{q_*} \left[\left(\frac{p(x)}{q_*(x)} \right)^2 p(y|x)^2 \right] &= \int \frac{p^2(x)}{q_*^2(x)} p(y|x)^2 q_*(x) dx \\ &= \int \frac{p^2(x)}{q_*(x)} p(y|x)^2 dx \\ &= \int \frac{p^2(x)}{p(x|y)} p(y|x)^2 dx \\ &= \int \frac{p^2(x)p(y)}{p(y|x)p(x)} p(y|x)^2 dx \\ &= p(y) \int p(x)p(y|x) dx \\ &= p(y)^2. \end{aligned}$$

Plugging this back into the above variance expression $\text{var}_{q_*}[p_{\text{IS}}^N(y)]$, we obtain

$$\text{var}_{q_*}[p_{\text{IS}}^N(y)] = \frac{1}{N} (p(y)^2 - p(y)^2) = 0.$$

It can be seen that we can achieve zero variance, but as we mentioned before, this required us to know the posterior density.

Exercise 3.10. In this exercise, we will use our toy Gaussian model to perform Bayesian inference:

$$\begin{aligned} p(x) &= \mathcal{N}(x; \mu_0, \sigma_0^2), \\ p(y_i|x) &= \mathcal{N}(y_i; x, \sigma^2), \end{aligned}$$

where $i = 1, \dots, M$, i.e., y_i are conditionally independent given x . In the first implementation of this exercise, you can set $\mu_0 = 0$, $\sigma^2 = 1$ and $\sigma_0^2 = 1$, but you should also explore other parameters to gain intuition.

1. First simulate $M = 1000$ data points from this model. In particular, you should perform $X \sim p(x)$ and fixing the sample $X = x$, and then $y_i \sim p(y_i|x)$ for $i = 1, \dots, M$.
2. For these particular datapoints, compute the *true* posterior mean estimate using the expression μ_p in Example 3.6. This is the true posterior mean estimate for this model.
3. Now we will use SNIS to estimate the posterior mean. For this, we will use the following proposal

$$q(x) = \mathcal{N}(x; \mu_q, \sigma_q^2),$$

In the first implementation, you can set $\mu_q = 0$ and $\sigma_q^2 = 1$, but you should also explore other parameters to gain intuition. Implement the SNIS estimator for this model. Note that your unnormalized posterior is

$$\bar{p}(x|y_{1:n}) = p(y_{1:n}|x)p(x) = \prod_{i=1}^M p(y_i|x)p(x),$$

as given in Exercise 3.6. You should compute the weights in the log-domain. Compare the posterior mean estimate when you compute the weights without the log-trick and with the log-trick. You should observe that the log-trick is necessary for this model.

4. Finally compute effective sample size (ESS) and vary your N to see how it affects the ESS. Plot the ESS w.r.t. N .

Exercise 3.11. In this example, we will work on $\mathbb{R}^2$. Let us assume the following “banana” prior

$$p(x) \propto \exp(-x_1^2/10 - x_2^2/10 - 2(x_2 - x_1^2)^2),$$

where $x \in \mathbb{R}^2$. We would like to then use the following likelihood:

$$p(y|x) = \mathcal{N}(y; Hx, \sigma^2 I),$$

where $H = [1, 0]$ and $y \in \mathbb{R}$. Now assume that we are given $y = 1$ and $\sigma = 0.05$ (note that this is standard deviation). Our goal is to sample from $p(x|y)$.

1. First visualise the prior $p(x)$ using the following code:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def bar_p(x):
5     return np.exp(-x[0]**2/10 - x[1]**2/10 - 2 * (x[1] - x[0]**2)**2)
6
7 x_bb = np.linspace(-4, 4, 100)
8 y_bb = np.linspace(-2, 6, 100)
9 X_bb, Y_bb = np.meshgrid(x_bb, y_bb)
```

```

10 # evaluate barp on this grid
11 Z_bb = np.zeros((100, 100))
12 for i in range(100):
13     for j in range(100):
14         Z_bb[i, j] = bar_p([X_bb[i, j], Y_bb[i, j]])
15 # plot barp
16 plt.contourf(X_bb, Y_bb, Z_bb, 100, cmap='RdBu')
17 plt.show()

```

You will see that this gives you a 2D (unnormalised) density. Now try to interpret our task. We assume that $y = 1$ with $\sigma = 0.05$. By just looking at the prior and the likelihood, try to guess how the posterior should look like (before proceeding).

2. Now we will use SNIS to estimate the posterior mean. For this, we will use the following proposal

$$q(x) = \mathcal{N}(x; \mu_q, \sigma_q^2 I),$$

Note that this is also defined on $\mathbb{R}^2$ and you can set $\mu_q = [0, 0]$ and $\sigma_q^2 = 1$. Compute the posterior mean using SNIS (do not forget using log-prior and log-likelihood and performing the log-trick) and compute the ESS (choose varying N . Start with $N = 1000$).

3. Now using the idea of importance sampling resampling (Section 4.6.2) in lecture notes, resample N samples from this weighted sample. Recall that SNIS will result in a set of weights and samples $\{\bar{w}_i, x_i\}_{i=1}^N$. You should resample from this set to obtain a new set of *resampled* samples $\{\bar{x}_i\}_{i=1}^N$. Then scatter these samples to plot against the prior using the following code:

```

1 # plot resampled samples
2 # x_res are the resampled samples (note that samples are 2D)
3 x_bb = np.linspace(-4, 4, 100)
4 y_bb = np.linspace(-2, 6, 100)
5 X_bb, Y_bb = np.meshgrid(x_bb, y_bb)
6 Z_bb = np.zeros((100, 100))
7 for i in range(100):
8     for j in range(100):
9         Z_bb[i, j] = bar_p([X_bb[i, j], Y_bb[i, j]])
10 plt.contourf(X_bb, Y_bb, Z_bb, 100, cmap='RdBu')
11 plt.scatter(x_res[0, :], x_resampled[1, :], s=10, c='white')
12 plt.show()

```

Now interpret the result and compare it to your educated guess you have done in Part 1.

3.5 IMPLEMENTATION, ALGORITHMS, DIAGNOSTICS

When implementing the IS or SNIS, there are several numerical considerations that need to be taken into account. Especially for SNIS, where the weight normalisation takes place, several numerical problems can arise for complex distributions that would prevent us from implementing

them successfully.

3.5.1 COMPUTING WEIGHTS

In the case of SNIS, we have stated that the weights are computed as

$$\bar{w}_i = \frac{W(X_i)}{\sum_{i=1}^N W(X_i)},$$

where $W(x) = \frac{\bar{p}(x)}{q(x)}$. However, this formula can be numerically ill-behaved for complex distributions. To mitigate this, the weighting step is implemented as follows. We first compute log *unnormalised* weights:

$$\log W_i = \log \bar{p}(X_i) - \log q(X_i).$$

However, directly applying exponentiation and normalisation can also lead to numerical problems. Note that, we will normalise these weights (e.g. multiplying them with a constant does not change the result), we can use this to our advantage. A common numerical trick is to subtract the maximum log weight from all weights:

$$\log \tilde{W}_i = \log \bar{p}(X_i) - \log q(X_i) - \max_{i=1, \dots, N} \log W_i.$$

This ensures that the maximum weight is 0 and all other weights are negative. We can now exponentiate the weights and normalise them:

$$\bar{w}_i = \frac{\exp(\log \tilde{W}_i)}{\sum_{i=1}^N \exp(\log \tilde{W}_i)}.$$

Note that, this does not change the computation, just done for numerical stability.

Exercise 3.12. Consider a computed log-weight vector

1 `logW = [1000, 1001, 999, 1002, 950]`

These are computed log-weights which are, for various reasons, often the only available quantities in practice (people implement quadratics, rather than Gaussians, for example). Implement the naive normalisation procedure

$$w_i = \frac{\exp(\log W_i)}{\sum_{i=1}^N \exp(\log W_i)}.$$

Implement the trick introduced in Sec. 4.4.1 in lecture notes and verify that the latter computation is stable.

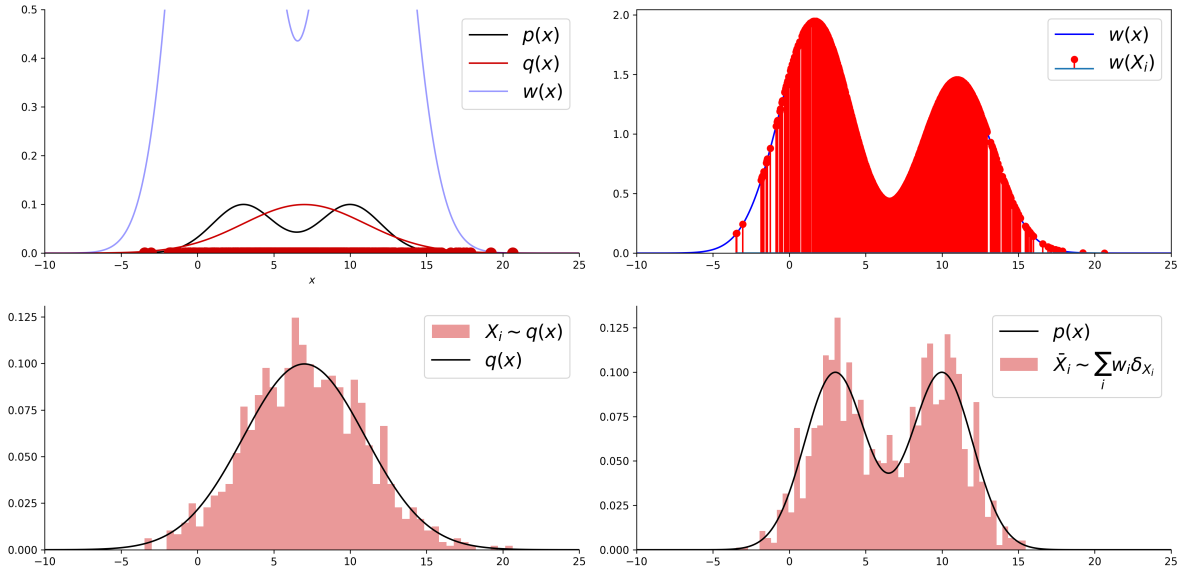


Figure 3.11: Top left shows the target density, the proposal, and the weight function. Top right shows the samples with their respective weights. Bottom left shows that these samples are indeed approximately distribution w.r.t. $q(x)$ (just with attached weights). Bottom right shows that we can resample these existing samples to obtain a new set of samples $\tilde{X}_i$ that are distributed (approximately) according to $p(x)$.

3.5.2 SAMPLING IMPORTANCE RESAMPLING

We can also use the SNIS estimator as a sampler (Robert and Casella, 2004). Recall that, the SNIS estimator provides us an estimator of the distribution $p_*(x)$ as

$$p_*(x)dx \approx \tilde{p}_*^N(dx) = \sum_{i=1}^N \bar{w}_i \delta_{X_i}(dx).$$

This $\tilde{p}_*^N$ can be seen as a weighted distribution. By drawing samples from this distribution, we may also approximately sample from $p_*(x)$ (recall that IS based ideas here are just introduced for integration so far). We can then draw⁴

$$k \sim \text{Discrete}(\bar{w}_1, \dots, \bar{w}_N),$$

and set new samples

$$\tilde{X}_i = X_k.$$

This amounts to *resampling* the existing samples w.r.t. their weights. A demonstration of this idea can be seen from Figure 3.11.

3.5.3 DIAGNOSTICS FOR IMPORTANCE SAMPLING

It is important to have a good intuition and diagnostic tools to understand the performance of the IS and SNIS estimators. We first start with the effective sample size (ESS).

⁴Note that here the weights $\bar{w}_i$ are normalised. Even in the basic IS case, we need to normalise weights (**just for resampling**) as they do not naturally sum up to one.

Definition 3.1 (Effective Sample Size). *To measure the sample efficiency, one measure that is used in the literature is the effective sample size (ESS) which is given by*

$$\text{ESS}_N = \frac{1}{\sum_{i=1}^N \bar{w}_i^2},$$

for the SNIS estimator.

In order to see the meaning of the ESS, consider the case where $\bar{w}_i = 1/N$ where we have an equally weighted sample. This means all samples are equally considered and in this case we have $\text{ESS}_N = N$. On the other hand, if we have a sample X_i where $\bar{w}_i = 1$ and, hence, $\bar{w}_j = 0$ for every $j \neq i$, we obtain $\text{ESS}_N = 1$. This means, we *effectively* have one sample which is the goal of the estimator. ESS is used to measure importance samplers and importance sampling-based estimators in the literature (Elvira et al., 2018). Note that the ESS_N takes values between 1 and N , i.e., $1 \leq \text{ESS}_N \leq N$.

3.5.4 MIXTURE IMPORTANCE SAMPLING

Sometimes the target density $p(x)$ can be multimodal, therefore it is beneficial to use mixture densities as proposals (Owen, 2013). We have seen in previous chapters how to sample from a mixture. Let us define a proposal

$$q_\alpha(x) = \sum_{k=1}^K \alpha_k q_k(x),$$

where $\alpha_k \geq 0$ and $\sum_{k=1}^K \alpha_k = 1$. In this version of the method, we just sample from the mixture proposal $X_i \sim q_\alpha(x)$ and then, given an unnormalised target $\bar{p}$, compute the importance weights as

$$\bar{w}_i = \frac{W(X_i)}{\sum_{i=1}^N W(X_i)},$$

where

$$W(X_i) = \frac{\bar{p}(X_i)}{\sum_{k=1}^K \alpha_k q_k(X_i)}.$$

The computational concerns may arise in this situation too, as the denominator as a sum of densities and its log can be tricky to compute. In these cases, we can use the log-sum-exp trick to compute the log of the denominator.

BIBLIOGRAPHY

- Agapiou, Sergios; Papaspiliopoulos, Omiros; Sanz-Alonso, Daniel; and Stuart, Andrew M. 2017. *Importance sampling: Intrinsic dimension and computational cost*. In *Statistical Science*, pp. 405–431. Cited on p. 86.
- Akyildiz, Omer Deniz. March 2019. *Sequential and adaptive Bayesian computation for inference and optimization*. Ph.D. thesis, Universidad Carlos III de Madrid. Can be accessed from: <http://akyildiz.me/works/thesis.pdf>. Cited on pp. 63, 67, and 75.
- Akyildiz, Ömer Deniz and Míguez, Joaquín. 2021. *Convergence rates for optimised adaptive importance samplers*. In *Statistics and Computing*, vol. 31, no. 2, pp. 1–17. Cited on pp. 78 and 86.
- Branchini, Nicola. 2024. *Simple Monte Carlo is independent of dimension. Or is it?* https://branchini.fun/posts/monte_carlo_indep/. Blog post. Cited on p. 65.
- Devroye, Luc. 1986. *Non-Uniform Random Variate Generation*. Cited on p. 17.
- Elvira, Víctor; Martino, Luca; and Robert, Christian P. 2018. *Rethinking the effective sample size*. In *International Statistical Review*. Cited on p. 95.
- Kallenberg, Olav. 2021. *Foundations of Modern Probability*. Probability Theory and Stochastic Modelling. Springer Cham, 3rd edn. ISBN 978-3-030-61871-1. doi:<https://doi.org/10.1007/978-3-030-61871-1>. 1 b/w illustration. Cited on p. 30.
- Martino, Luca; Luengo, David; and Míguez, Joaquín. 2018. *Independent random sampling methods*. Springer. Cited on pp. 18, 34, 35, and 39.
- Murphy, Kevin P. 2007. *Conjugate Bayesian analysis of the Gaussian distribution*. In *def*, vol. 1, no. 2σ2, p. 16. Cited on p. 6.
- Owen, Art B. 2013. *Monte Carlo theory, methods and examples*. Cited on p. 95.
- Robert, Christian P and Casella, George. 2004. *Monte Carlo statistical methods*. Springer. Cited on pp. 63 and 94.
- . 2010. *Introducing Monte Carlo methods with R*, vol. 18. Springer. Cited on p. 76.
- Walters, Peter. 2000. *An introduction to ergodic theory*, vol. 79. Springer Science & Business Media. Cited on p. 19.
- Yıldırım, Sinan. 2017. *Sabancı University IE 58001 Lecture notes: Simulation Methods for Statistical Inference*. Cited on p. 48.